Numerical Geometric Acoustics

Dissertation Defense

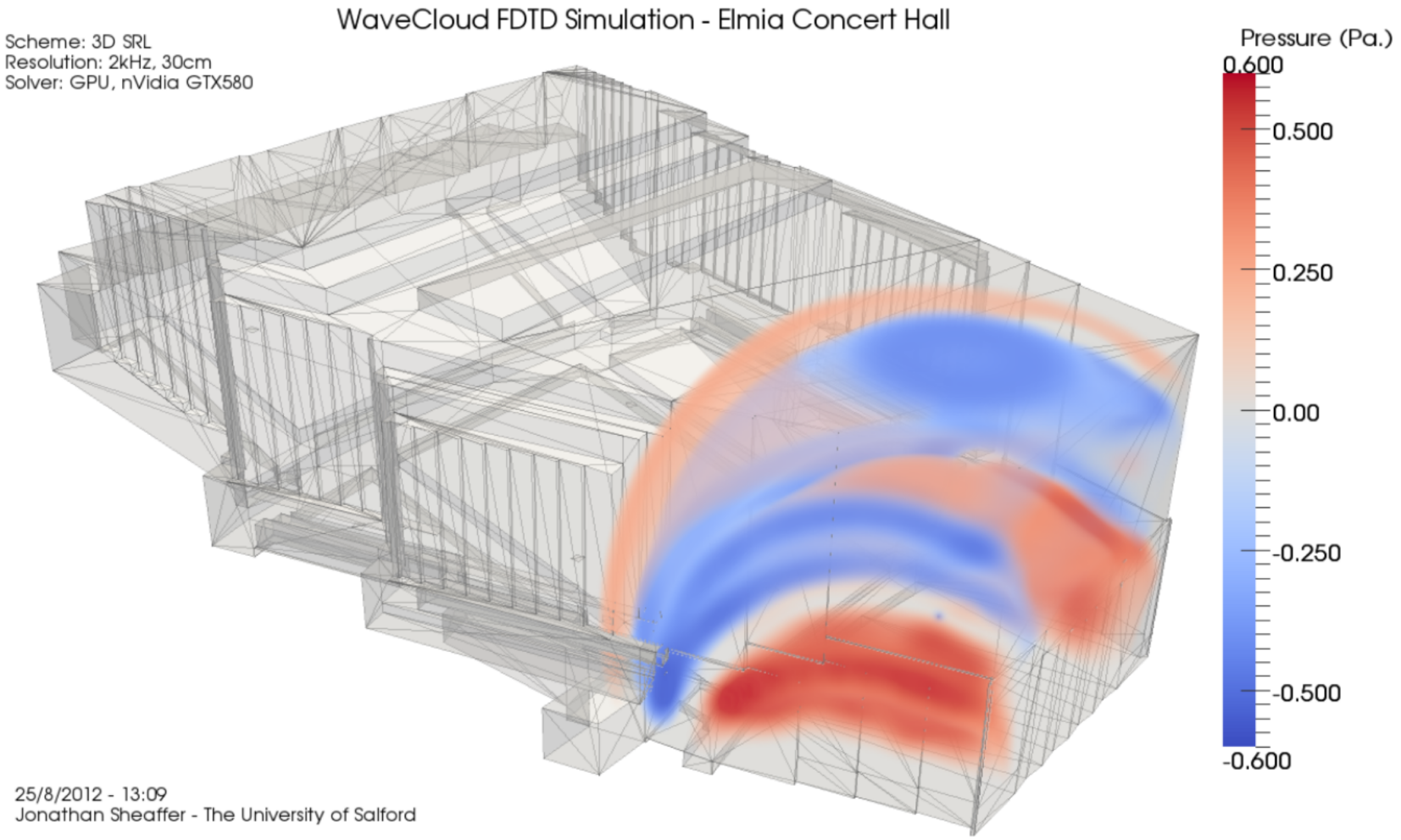
Samuel F. Potter

Advised by: Dr. Maria Cameron and Dr. Ramani Duraiswami

University of Maryland Department of Computer Science

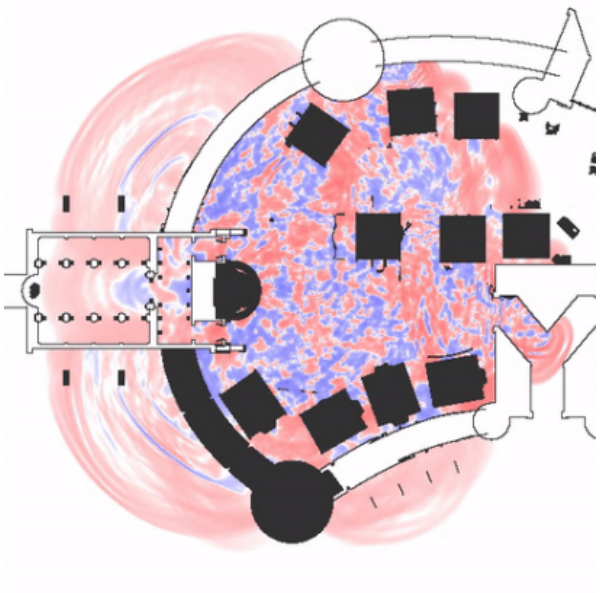
Introduction

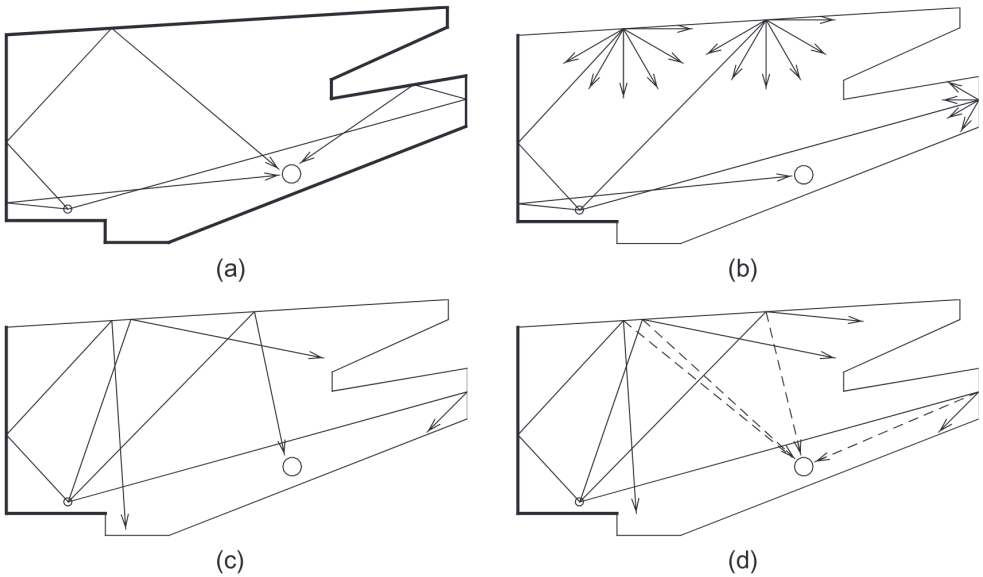
Introduction: computational room acoustics



Sheaffer, 2013.

Introduction: computational room acoustics





Numerical acoustics: solve wave equation for bandlimited impulse input. High-frequency ($>$2kHz) simulations expensive and impractical.

Geometric acoustics: use raytracing to assemble room impulse response. Handling diffraction effects with raytracing is complicated.

Left: `https://www.microsoft.com/en-us/research/project/project-triton`

Right: Kuttruff, 2016.

Introduction: PDEs for acoustics

Wave equation for pressure $p = p(\boldsymbol{x}, t)$:

$$p_{tt} = c^2 \Delta p$$

Time-harmonic solution for frequency ω:

$$p(\boldsymbol{x}, t) = e^{i\omega t} u_\omega(\boldsymbol{x})$$

Leads to the Helmholtz equation:

$$\Delta u_\omega + k^2 u_\omega = 0, \qquad k = \frac{\omega}{c}$$

where c is the speed of sound and k is the wavenumber.

Introduction: high-frequency wave propagation

Geometric optics ansatz ("WKB") for Helmholtz:

$$u_\omega(\boldsymbol{x}) \sim A_\omega(\boldsymbol{x}) e^{i\omega\tau(\boldsymbol{x})}$$

Leads to the eikonal equation:

$$\|\nabla\tau\| = 1/c = s, \qquad s \text{ is the "slowness"}$$

and the amplitude equation:

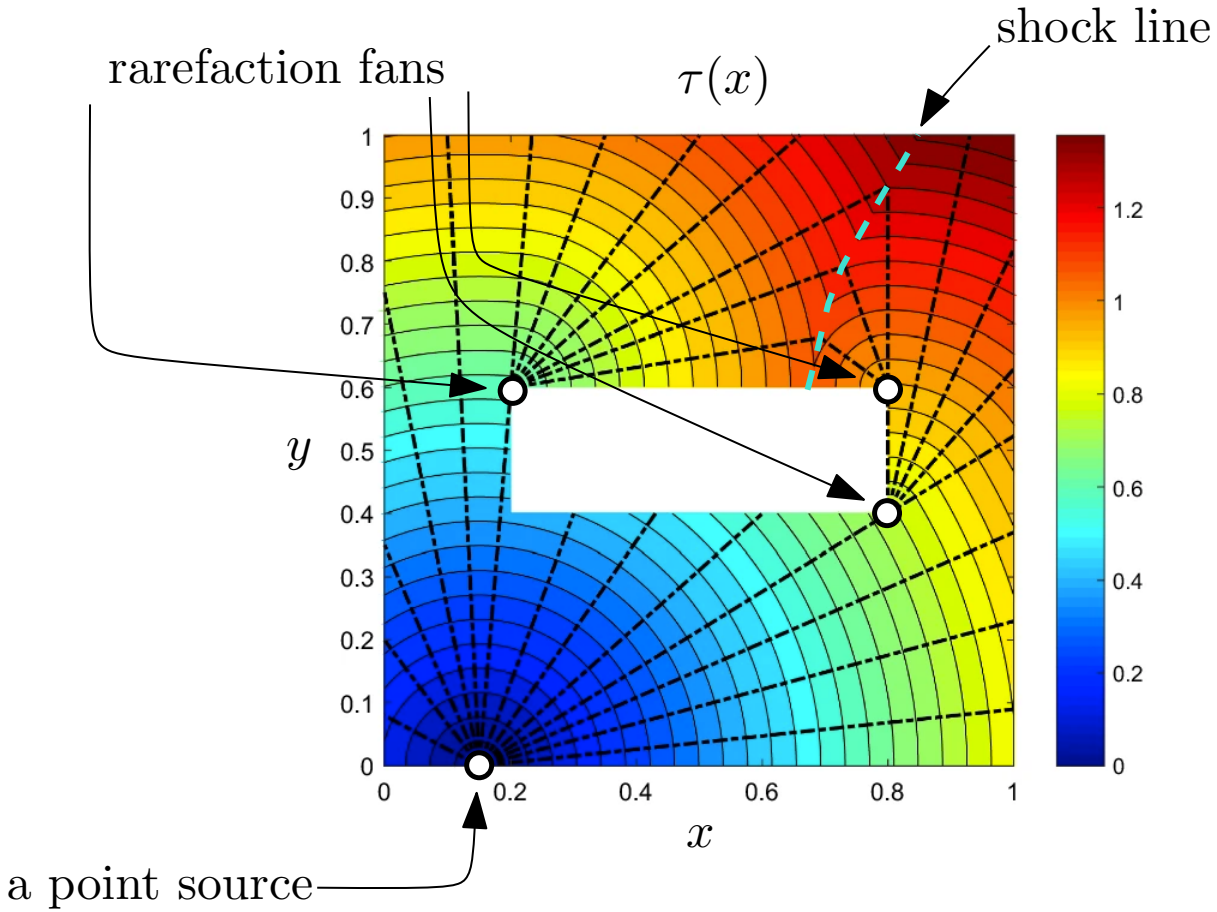
$$\nabla A_\omega \cdot \nabla\tau + 2A_\omega \Delta\tau = 0$$

High-frequency approximation of the Helmholtz equation:

$$|u_\omega - A_\omega e^{i\omega\tau}| = O(\omega^{-1})$$

Introduction: the eikonal equation, $\|\nabla\tau\| = 1/c$

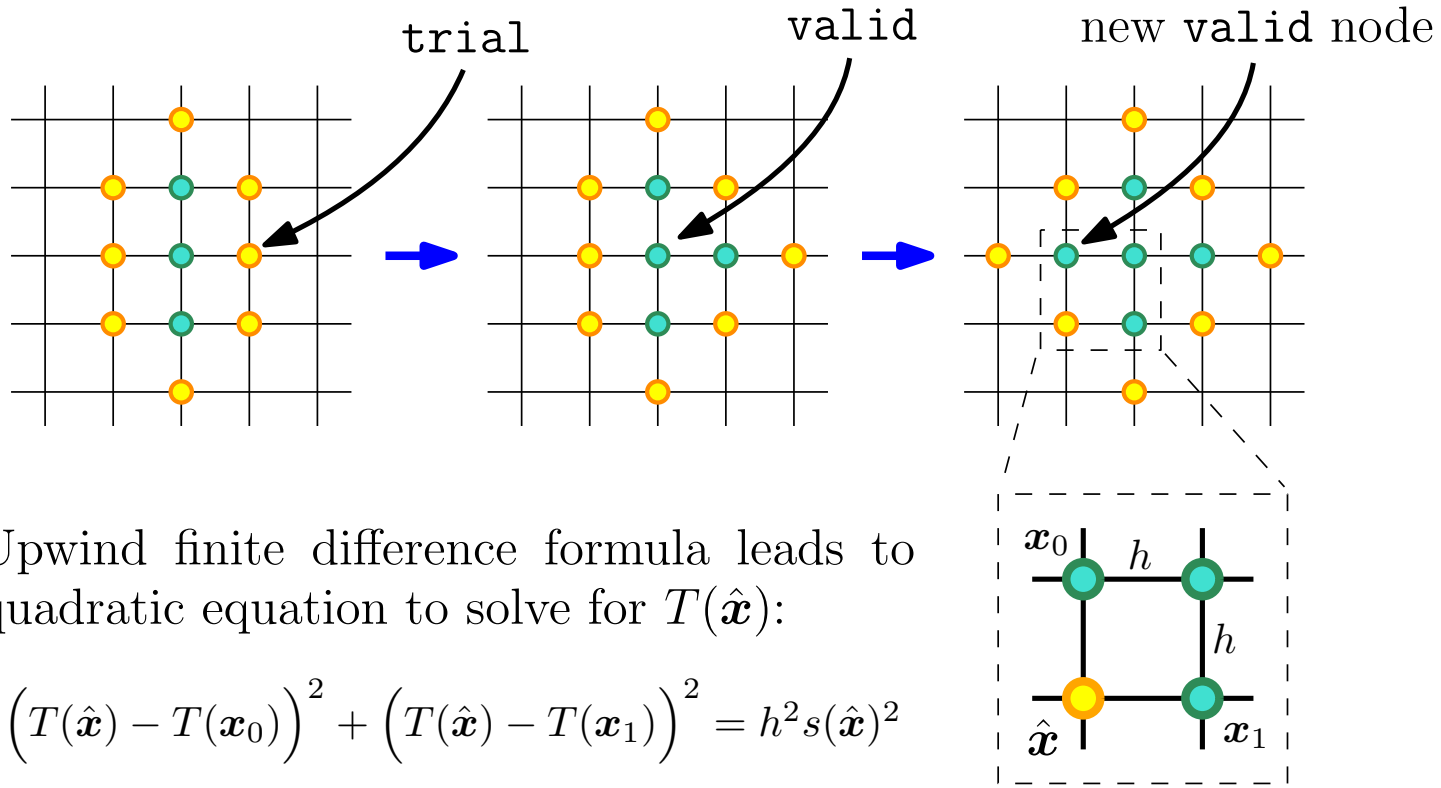
A PDE whose solution encodes only the first arrival time accounting for diffraction. Its characteristics are rays.



Source: Qi & Vladimirsky, 2019.

Introduction: the fast marching method

A single-pass, $O(h)$ accurate, upwind finite difference method for solving the eikonal equation based on Dijkstra's algorithm



Upwind finite difference formula leads to quadratic equation to solve for $T(\hat{\boldsymbol{x}})$:

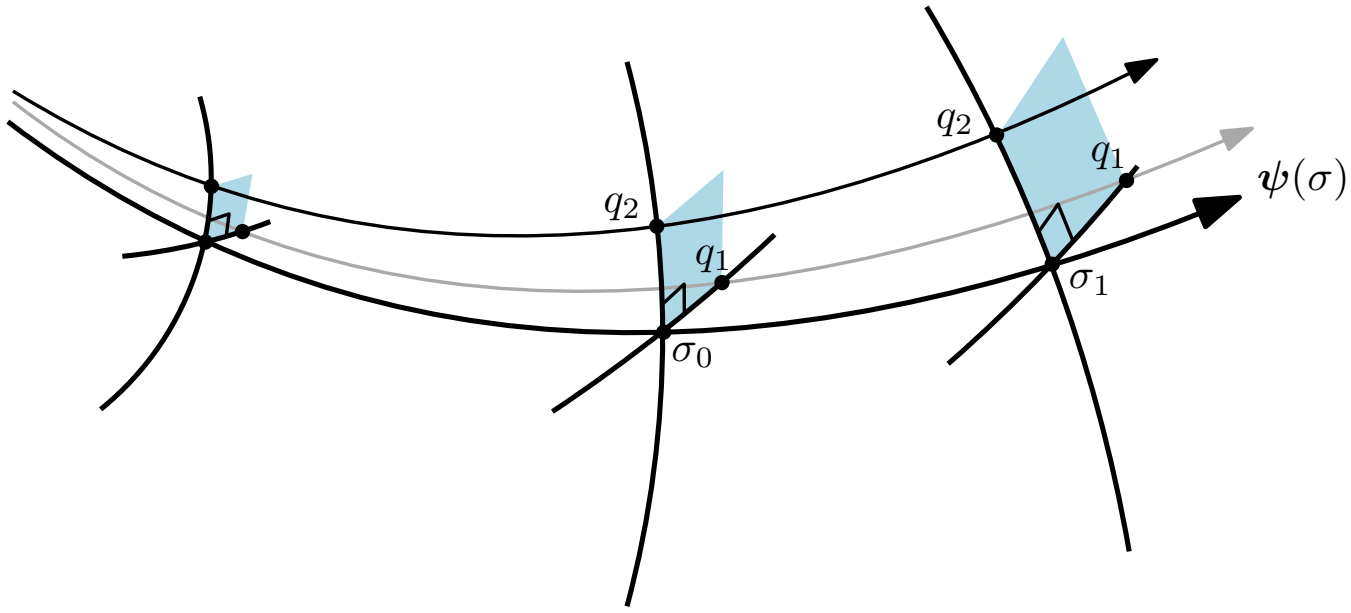
$$\Big(T(\hat{\boldsymbol{x}}) - T(\boldsymbol{x}_0)\Big)^2 + \Big(T(\hat{\boldsymbol{x}}) - T(\boldsymbol{x}_1)\Big)^2 = h^2 s(\hat{\boldsymbol{x}})^2$$

All `trial` nodes ordered in a priority queue by current T value

Sethian, 1996.

Introduction: the amplitude equation

Using **paraxial raytracing**, the amplitude can be computed in terms of the local geometric spreading of a ray tube:



The amplitude along a ray ψ satisfies:

$$\alpha(\psi(\sigma_1)) = \alpha(\psi(\sigma_0))\sqrt{\frac{s(\psi(\sigma_1))}{|\det(Q(\sigma_1))|}}, \qquad Q = \begin{bmatrix} q_1 & q_2 \end{bmatrix}$$

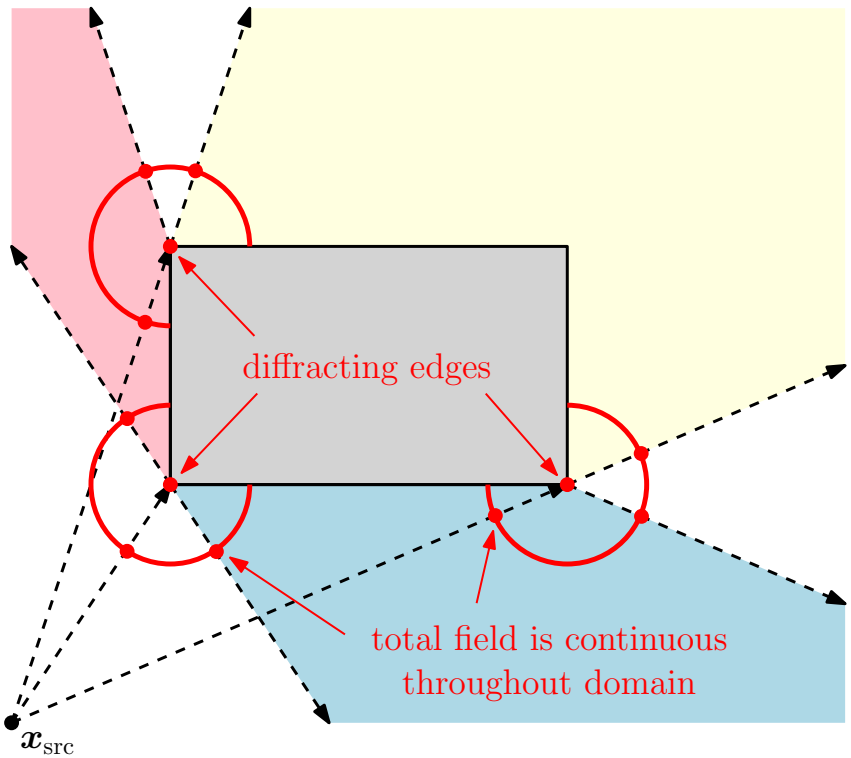
Popov, 1978.

Introduction: the geometric theory of diffraction

Multiple arrival high-frequency ansatz:

$$u_\omega(\boldsymbol{x}) \sim \sum_{\substack{k\text{th ray} \\ \text{through } \boldsymbol{x}}} A_\omega^{(k)}(\boldsymbol{x}) e^{i\omega\tau^{(k)}(\boldsymbol{x})}$$

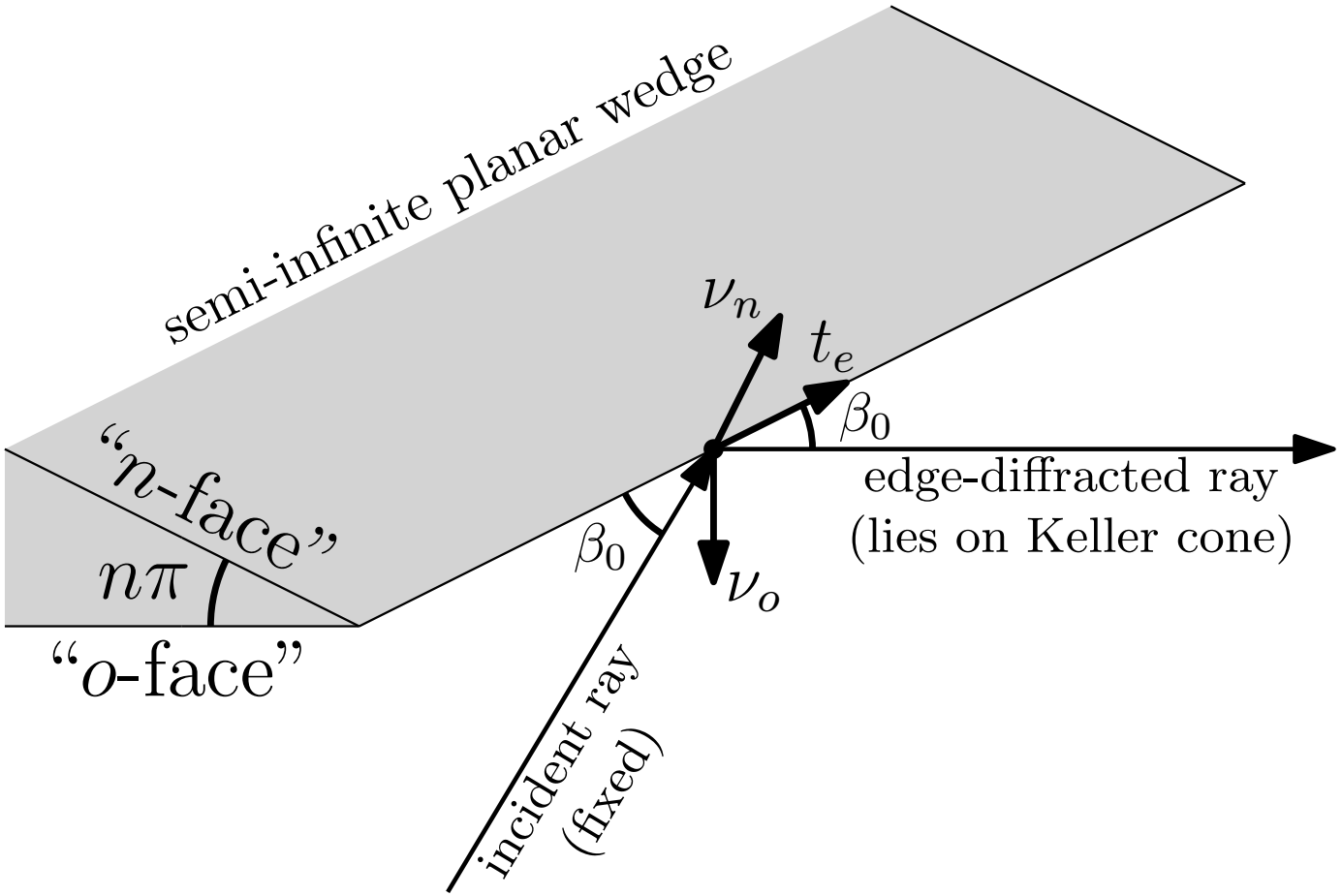
Generalized Fermat's principle with boundary constraints:



Keller, 1962.

Introduction: the uniform theory of diffraction

Uniform asymptotics which correct GTD's deficiencies:



Kouyoumjian, 1974.

Introduction: BCs for reflected and diffracted eikonal

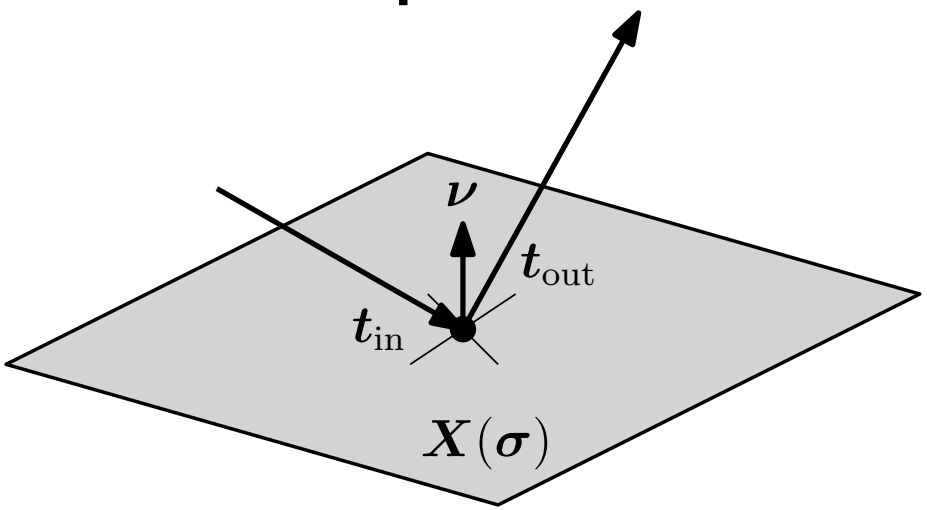
Boundary conditions for the reflected and diffracted eikonal:

$$\tau_{\text{in}}(\boldsymbol{X}(\boldsymbol{\sigma})) = \tau_{\text{out}}(\boldsymbol{X}(\boldsymbol{\sigma}))$$

Taking the gradient gives:

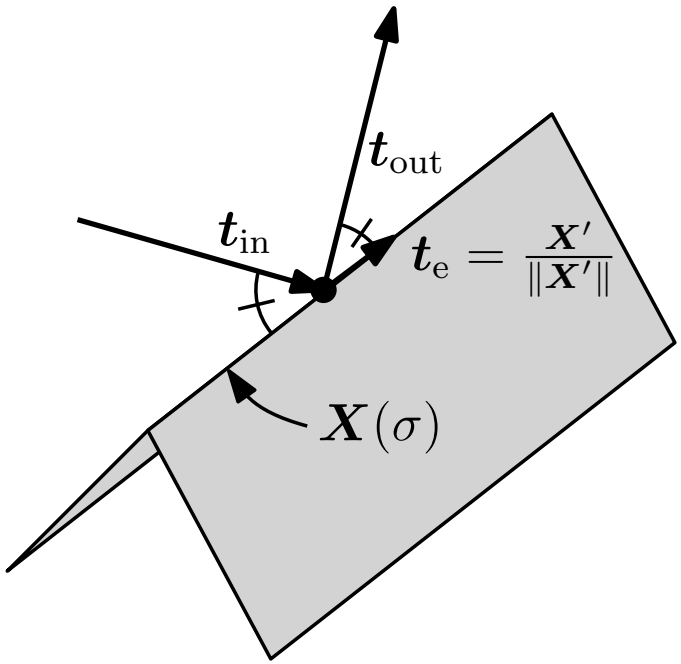
$$\nabla \boldsymbol{X}(\boldsymbol{\sigma})^\top \nabla \tau_{\text{in}}(\boldsymbol{X}(\boldsymbol{\sigma})) = \nabla \boldsymbol{X}(\boldsymbol{\sigma})^\top \nabla \tau_{\text{out}}(\boldsymbol{X}(\boldsymbol{\sigma}))$$

This leads to **specular reflection** and the **Keller cone**:



$$\boldsymbol{\nu} \in \text{null}(\boldsymbol{X})$$

$$\boldsymbol{t}_{\text{out}} = \left(\boldsymbol{I} - 2\boldsymbol{\nu}\boldsymbol{\nu}^\top\right)\boldsymbol{t}_{\text{in}}$$

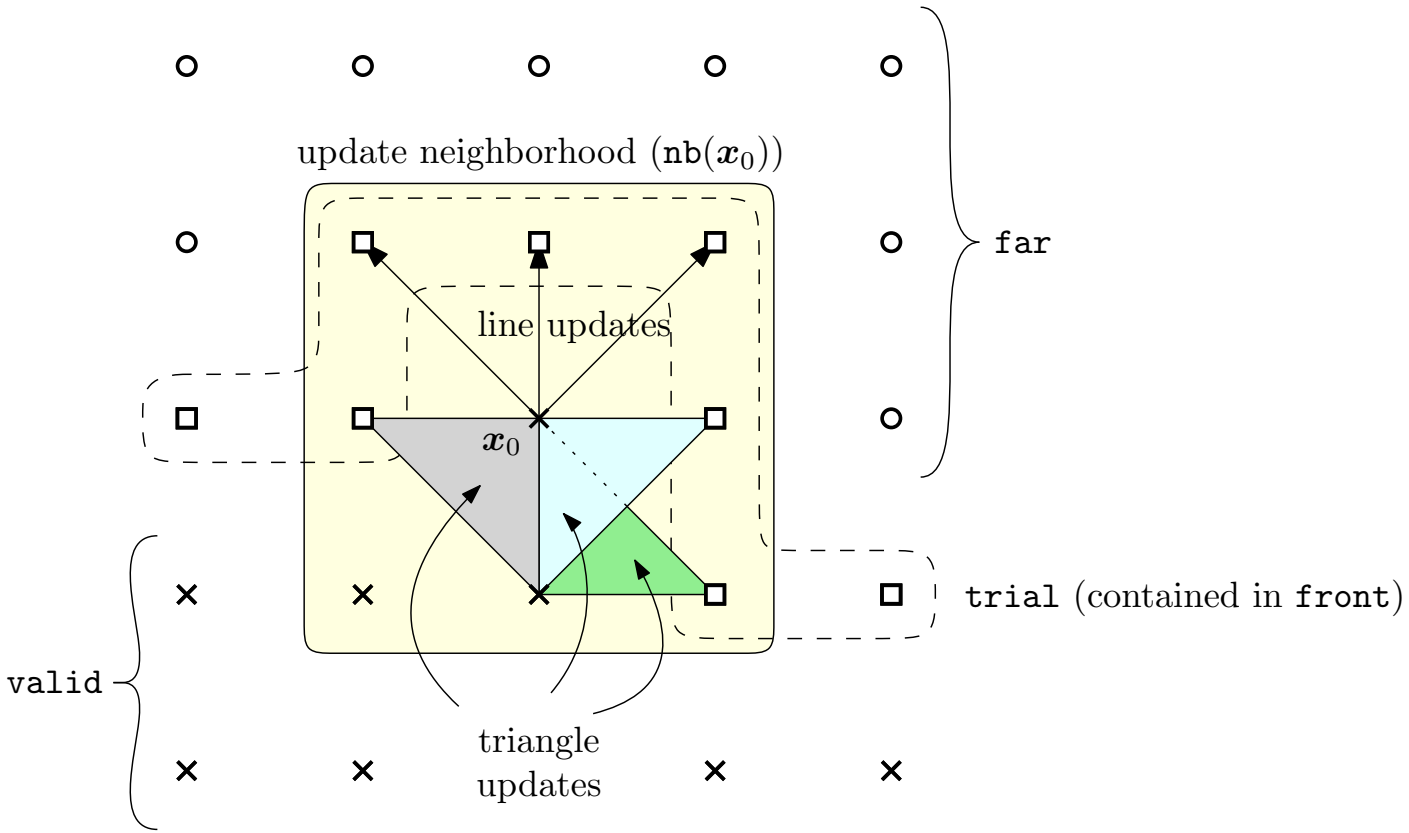


Ordered line integral methods (OLIMs) for solving the eikonal equation

OLIMs: semi-Lagrangian eikonal solvers

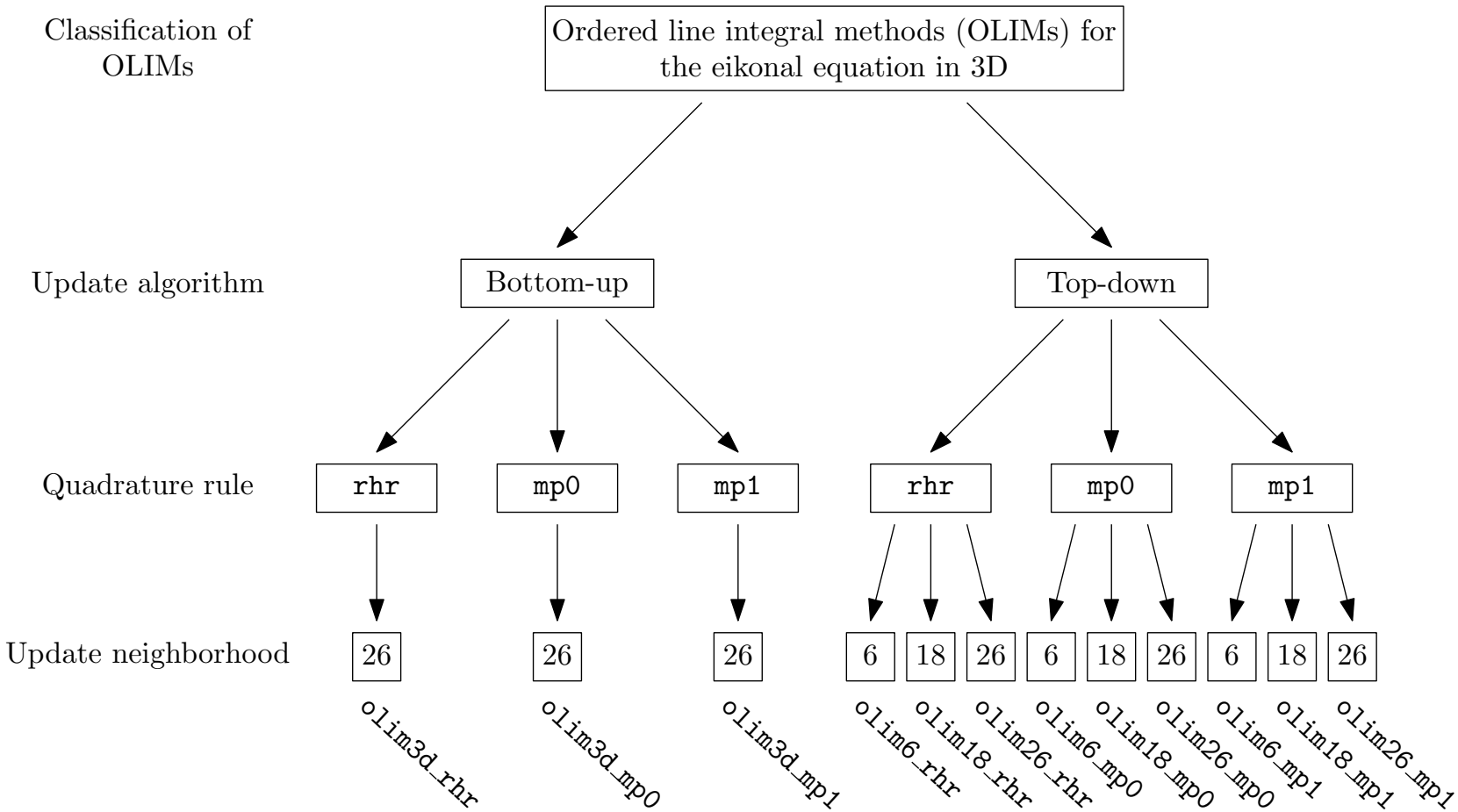
Apply Fermat's principle locally in a semi-Lagrangian fashion:

$$T(\hat{\boldsymbol{x}}) = \min_{\substack{\boldsymbol{\psi}:[0,1]\to\Omega \\ \boldsymbol{\psi}(0)=\boldsymbol{x} \\ \boldsymbol{\psi}(1)=\hat{\boldsymbol{x}}}} \left\{ T(\boldsymbol{x}) + \int_0^1 s(\boldsymbol{\psi}(\sigma))\|\boldsymbol{\psi}'(\sigma)\| d\sigma \right\}$$



Potter and Cameron, 2019.

OLIMs: parametrizing a family of solvers



Source: Potter and Cameron, 2019.

Potter and Cameron, 2019.

OLIMs: key takeaway

Both `mp0` and `mp1` are about an order of magnitude more accurate than `rhr` for smoothly varying c

We use the convergence theory for Newton's method (Kantorovich's theorem) to show that `mp0` and `mp1` differ in local error by at most $O(h^3)$

Each local update for a solver based on `mp1` requires us to solve a nonlinear constraint optimization problem

But, like `rhr`, a local update for `mp0` can be solved analytically using linear algebra

Combining `mp0` with a hierarchical update and rejecting updates using the KKT conditions leads to

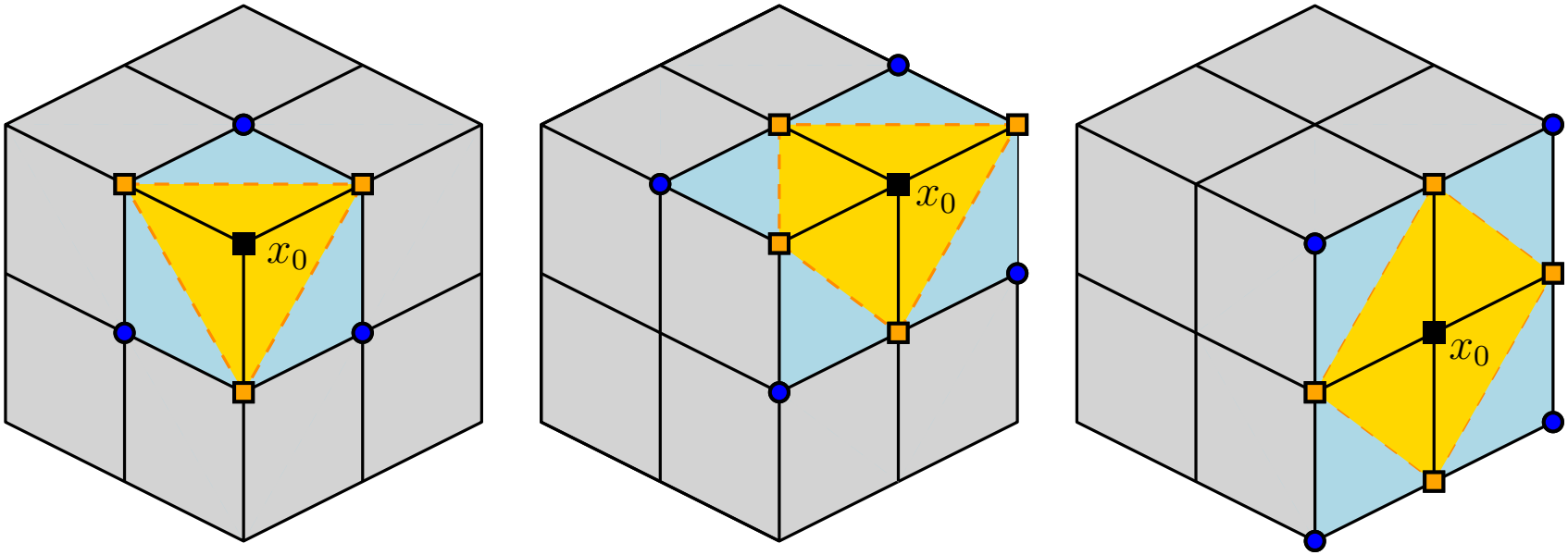
---

Potter and Cameron, 2019.

OLIMs: best choice of parameters

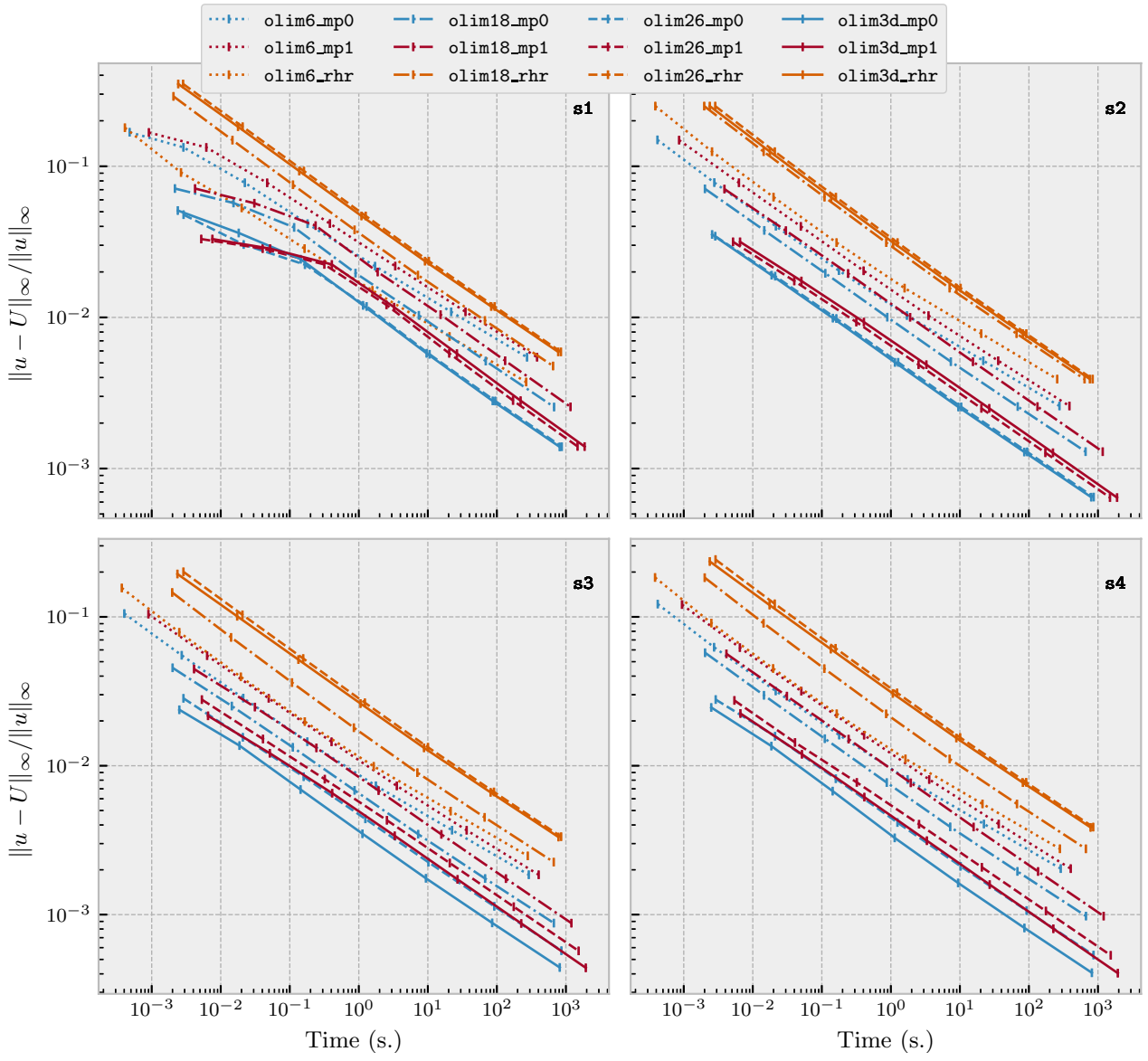
Best combination of parameters overall: `olim3d_mp0`

$$T(\hat{x}) = \min_{\lambda \in \Delta^d} \left\{ T(x_\lambda) + \frac{s(\hat{x}) + s(x_\lambda)}{2} \|\hat{x} - x_\lambda\| \right\}$$



Potter and Cameron, 2019.

OLIMs: numerical results

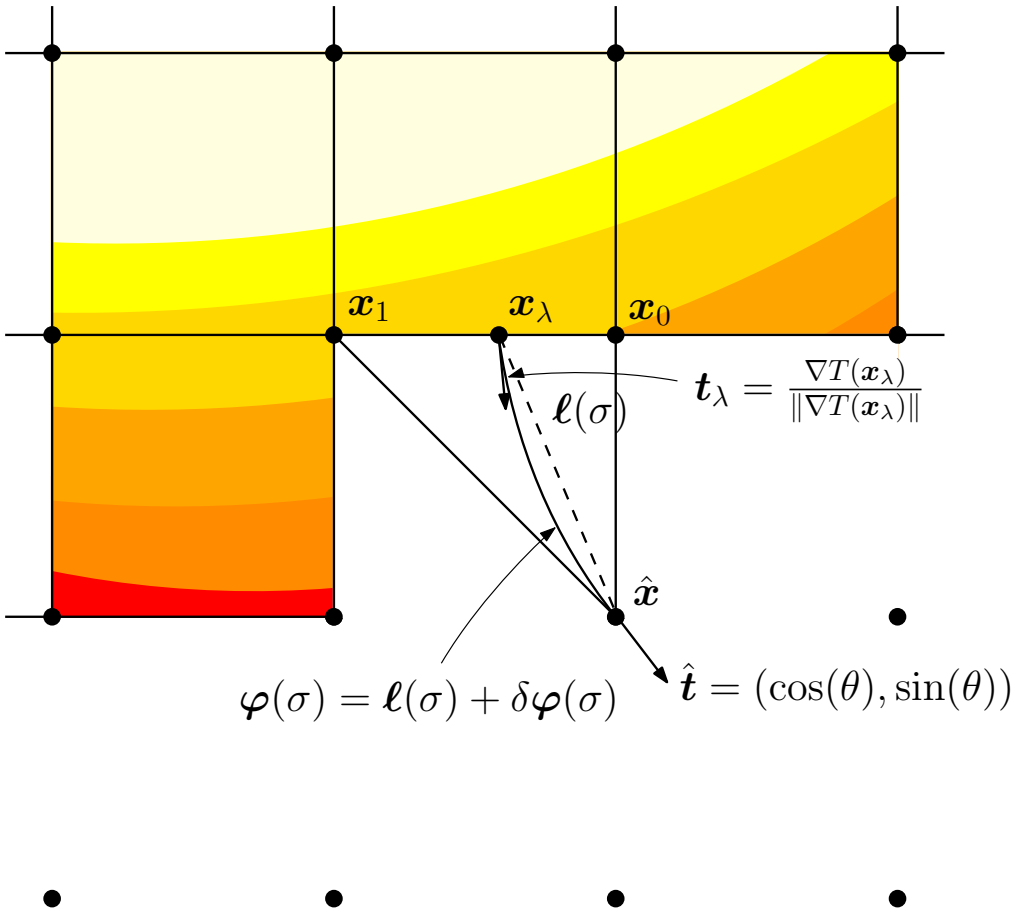


Potter and Cameron, 2019.

Jet marching methods (JMMs) in 2D

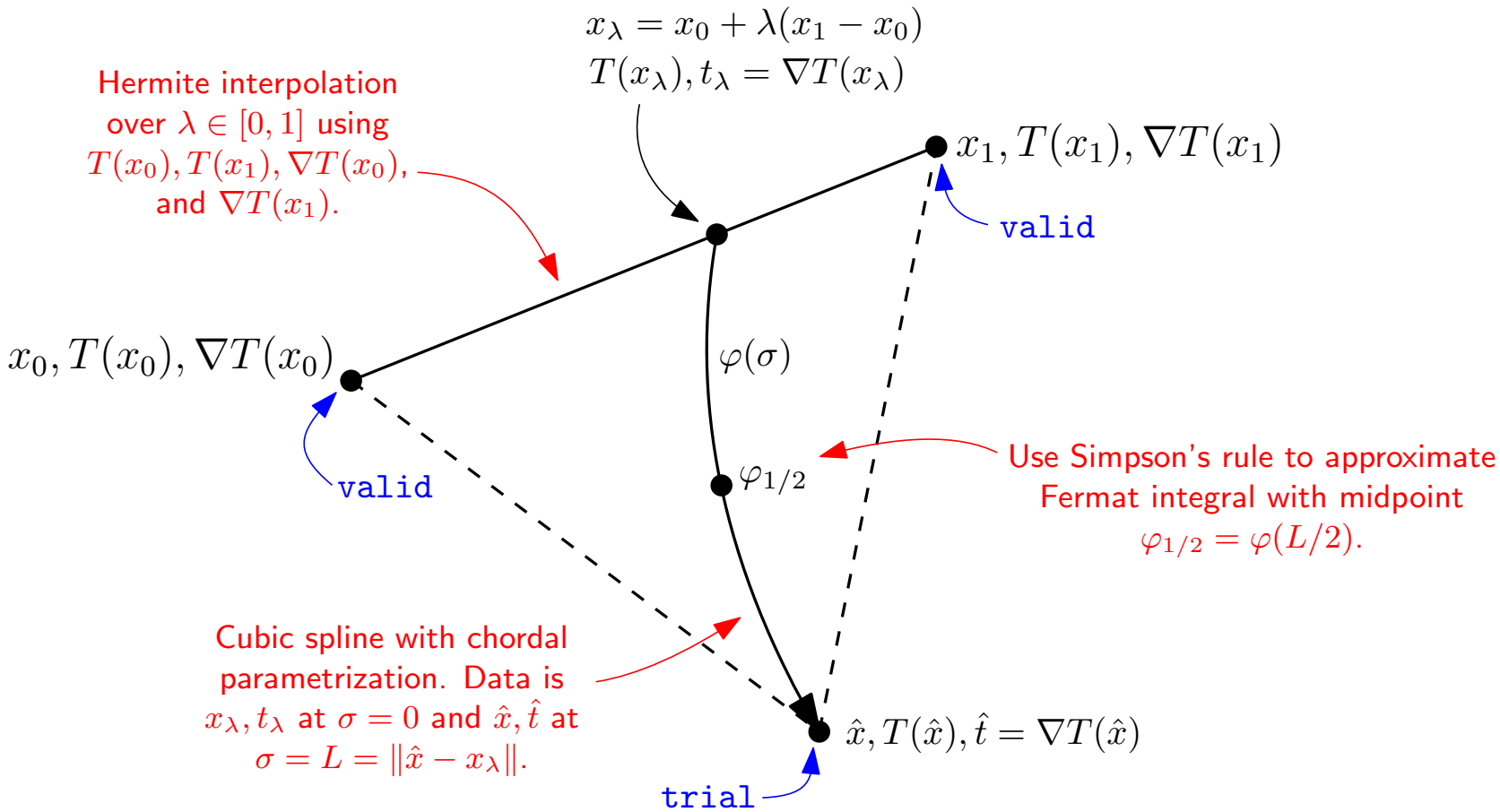
2D JMMs: overview

Approximate T and ψ using cubic polynomials, march the **jet** of the eikonal (coefficients of truncated Taylor polynomial):



Potter and Cameron, 2021.

2D JMMs: higher-order updates in 2D

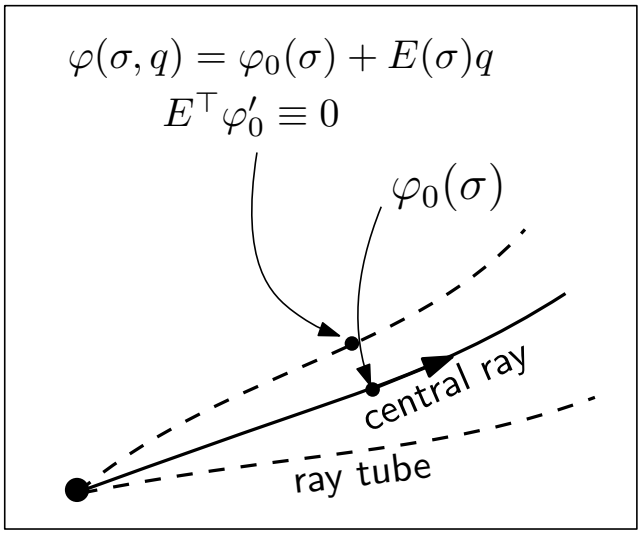


$$T(\hat{x}) = \min_{\substack{\hat{t}\in\mathbb{S}^1 \\ 0\le\lambda\le 1}} \left\{ \mathsf{T}(\lambda) + \frac{L}{6}\left(s(\hat{x}) + 4s(\varphi(\tfrac{L}{2}))\|\varphi'(\tfrac{L}{2})\| + s(x_\lambda) \right) \right\}$$

Potter and Cameron, 2021.

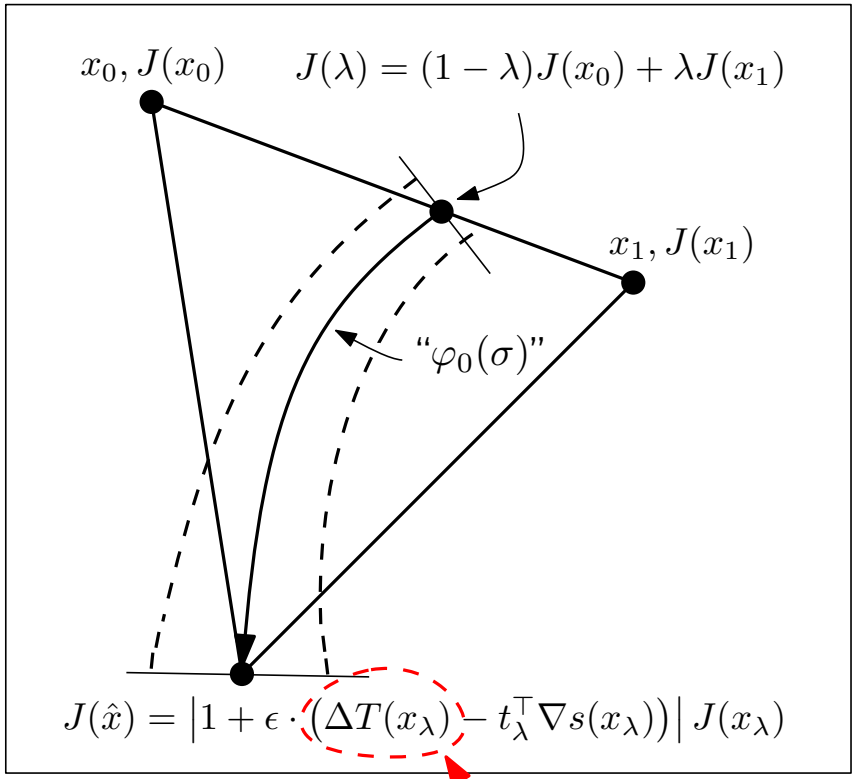
2D JMMs: marching the geometric spreading

Linearly approximate the geometric spreading J over $[\boldsymbol{x}_0, \boldsymbol{x}_1]$ and march it after parametrizing the local ray:



$$\begin{bmatrix} \dot{Q} \\ \dot{P} \end{bmatrix} = \begin{bmatrix} & c_0 I \\ \frac{-1}{c_0} \frac{\partial^2 c}{\partial q \partial q^\top}\Big|_{q=0} & \end{bmatrix} \begin{bmatrix} Q \\ P \end{bmatrix}$$

$$J = |\det Q|, \quad A(\phi_0(\sigma)) = \sqrt{\frac{c(\phi_0(\sigma))}{J(\phi_0(\sigma))}} A(\phi_0(0))$$

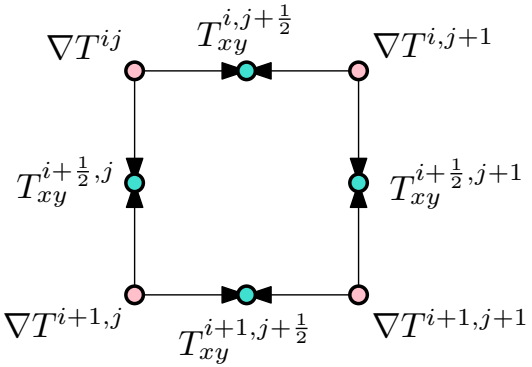


Need to know second partials of T at $x_\lambda \ldots$

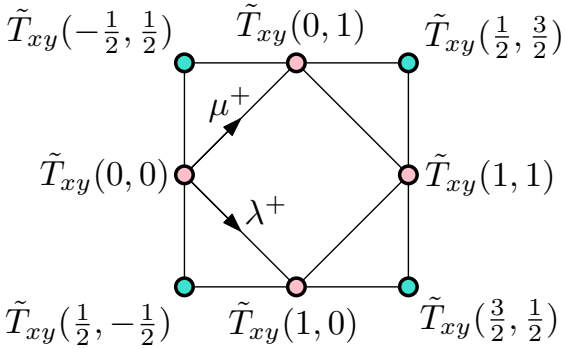
Potter and Cameron, 2021.

2D JMMs: cell marching

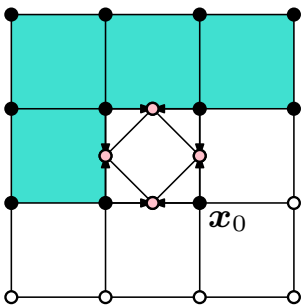
One way to march second partials (interpolation trick borrowed from Seibold et al., “Jet Schemes for Advection Problems”):



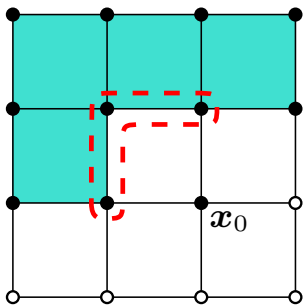
approximate T_{xy} at cell edge midpoints using central differences



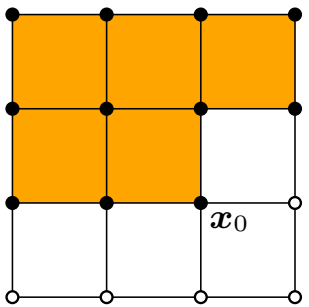
approximate T_{xy} at cell vertices using bilinear extrapolation



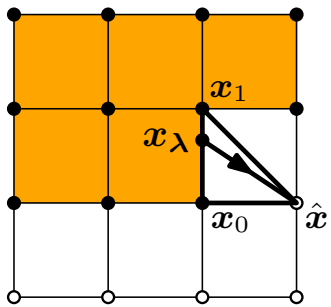
compute new T_{xy} values at corners of new `valid` cell



average T_{xy} values over adjacent `valid` cells



rebuild bicubics on affected `valid` cells



update neighboring `trial` nodes using triangle updates

Potter and Cameron, 2021.

2D JMMs: BCs for the geometric spreading

We march the geometric spreading using:

$$J(\hat{\boldsymbol{x}}) = \left|1 + \epsilon \cdot \left(\Delta T(\boldsymbol{x}_\lambda) - \boldsymbol{t}_\lambda \cdot \nabla s(\boldsymbol{x}_\lambda)\right)\right| \mathrm{J}(\boldsymbol{x}_\lambda),$$

where $\epsilon = \int_0^L c(\psi(\sigma))d\sigma$. We need BCs for J for a point source:

$$J(\boldsymbol{x}) \sim \|\boldsymbol{x}\| \text{ as } \boldsymbol{x} \to 0.$$

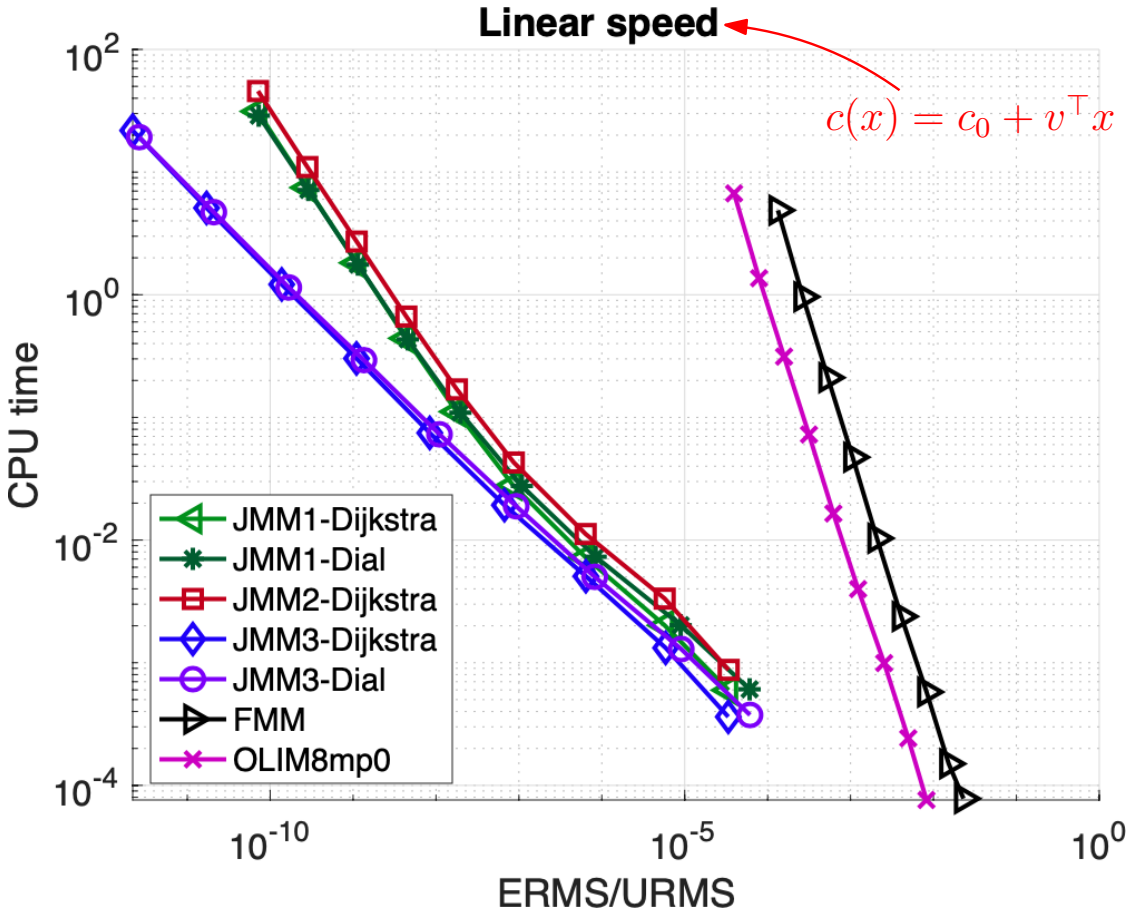
Afterwards, compute amplitude from:

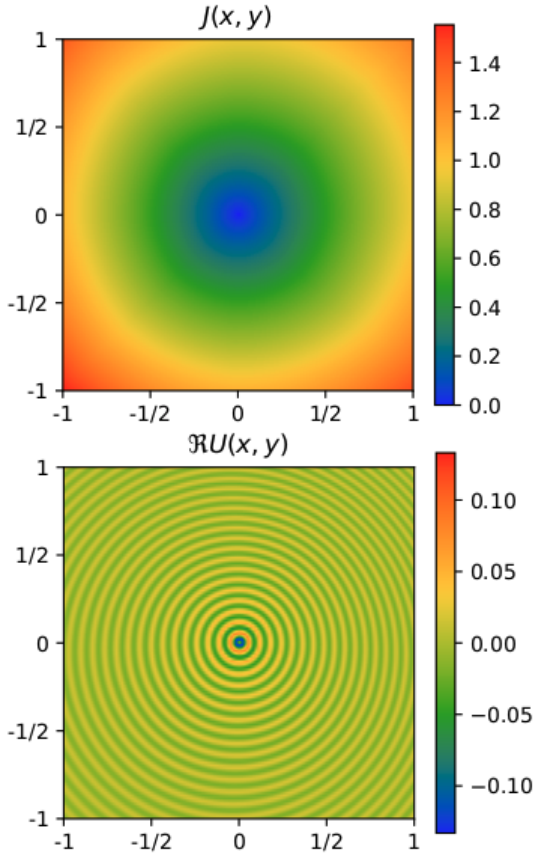
$$\alpha(\boldsymbol{x}) = \frac{e^{i\pi/4}}{2\sqrt{2\pi\omega}} \sqrt{\frac{c(\boldsymbol{x})}{J(\boldsymbol{x})}}.$$

This is specialized for the 2D case.

Potter and Cameron, 2021.

2D JMMs: numerical results

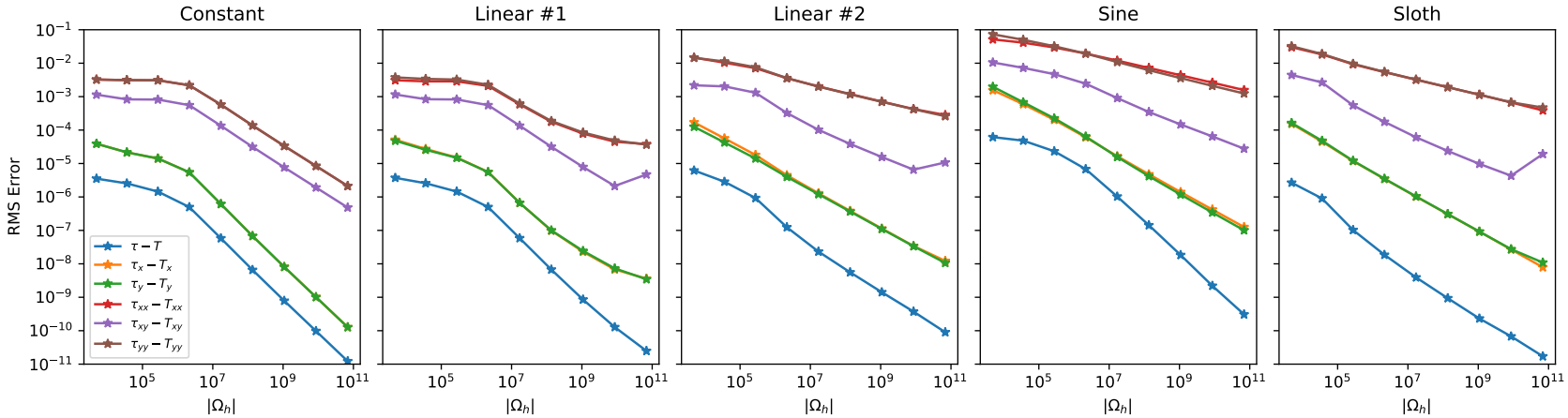




Potter and Cameron, 2021.

2D JMMs: numerical results

Results using cell marching:



Order of convergence ($O(h^p)$) for jets:

	$\tau - T$	$\tau_x - T_x$	$\tau_y - T_y$	$\tau_{xx} - T_{xx}$	$\tau_{xy} - T_{xy}$	$\tau_{yy} - T_{yy}$
Constant	3.09	3.11	3.11	2.01	2.05	2.01
Linear #1	2.99	2.43	2.40	1.39	2.01	1.39
Linear #2	2.10	1.76	1.72	0.77	1.25	0.77
Sine	2.91	1.80	1.89	0.73	1.31	0.80
Sloth	2.03	1.76	1.75	0.75	1.33	0.76

Potter and Cameron, 2021.

2D JMMs: preliminary theoretical results

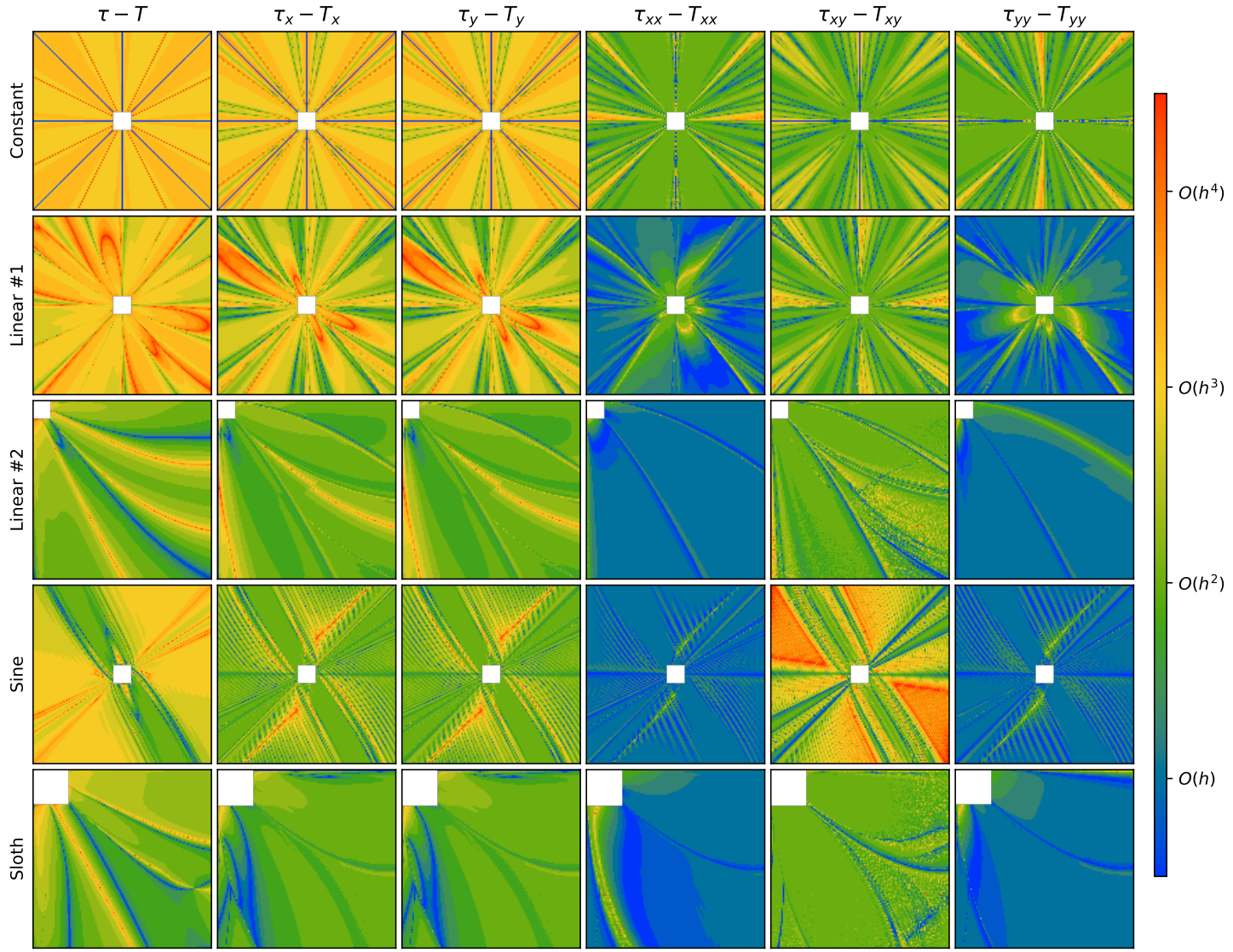
Theorem (`JMM3` consistency)

Let $\Omega \subseteq \mathbb{R}^2$, assume that $s \in C^2(\Omega)$, and let $(\hat{\boldsymbol{x}}, \boldsymbol{x}_1, \boldsymbol{x}_2)$ be an update triangle where $\hat{\boldsymbol{x}}$ is `trial`*, and $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are* `valid`*. Assume that no caustic is incident on the update triangle, and that the ray arriving at $\hat{\boldsymbol{x}}$ crosses the segment $[\boldsymbol{x}_1, \boldsymbol{x}_2]$. Then:*

$$|\tau(\hat{\boldsymbol{x}}) - T(\hat{\boldsymbol{x}})| = O(h^4), \qquad \|\nabla\tau(\hat{\boldsymbol{x}}) - \nabla T(\hat{\boldsymbol{x}})\| = O(h^2). \quad (1)$$

Potter and Cameron, 2021.

2D JMMs: global error



Potter and Cameron, 2021.

A JMM for an unstructured tetrahedron mesh

3D JMM: setup

Develop a JMM to solve the eikonal equation on a tetrahedron mesh in 3D

Assume that the speed of sound is constant (WLOG, $c \equiv 1$, so we compute the distance transform)

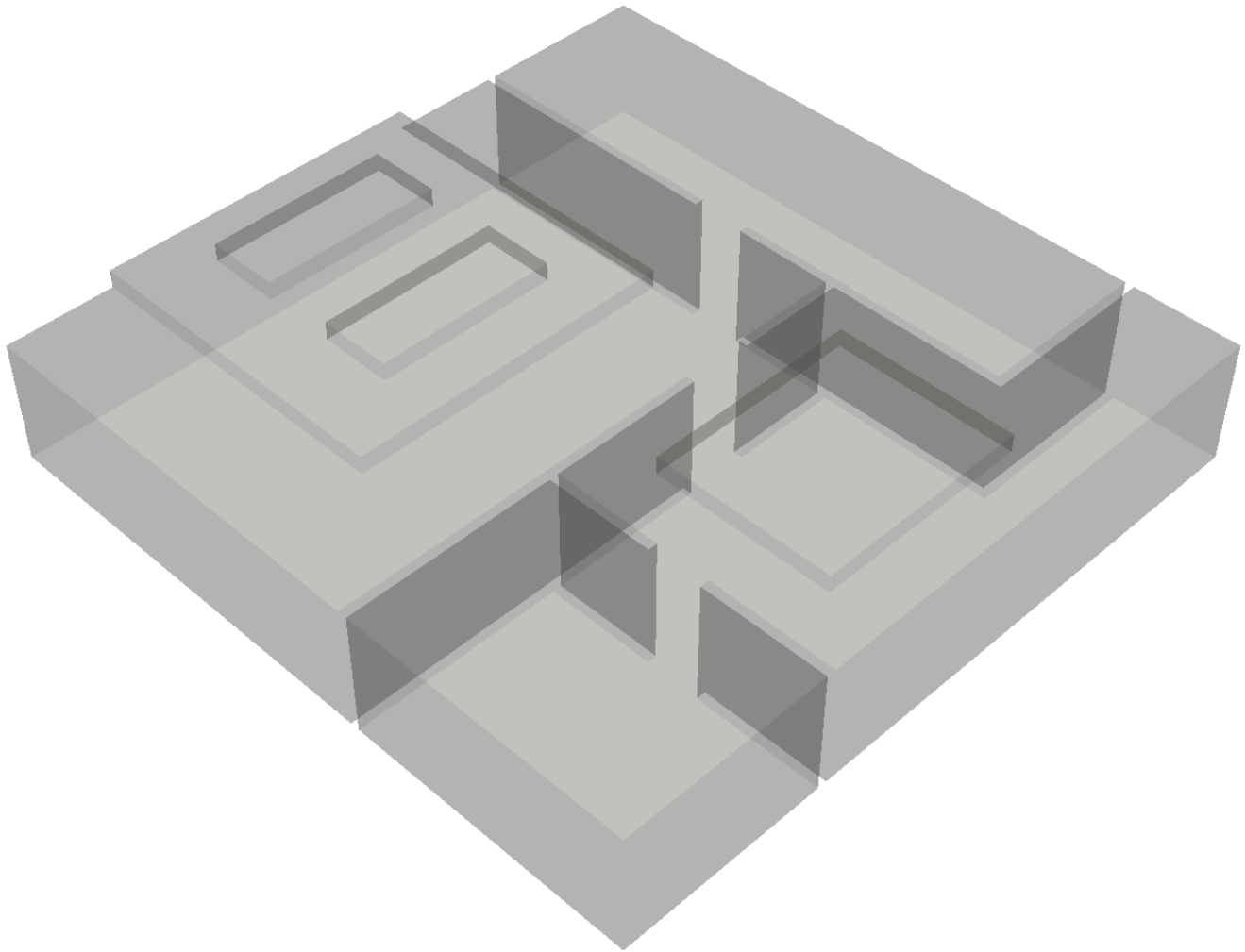
Goals: $O(N \log N)$ solver with at least $O(h^2)$ accurate T and ∇T, so that we can compute α consistently

Extend to spatially varying c later, since multiple arrival problem is already very complicated in 3D

Changes for varying c: have to solve slightly harder optimization problem for local updates, and have to deal with free-space caustics

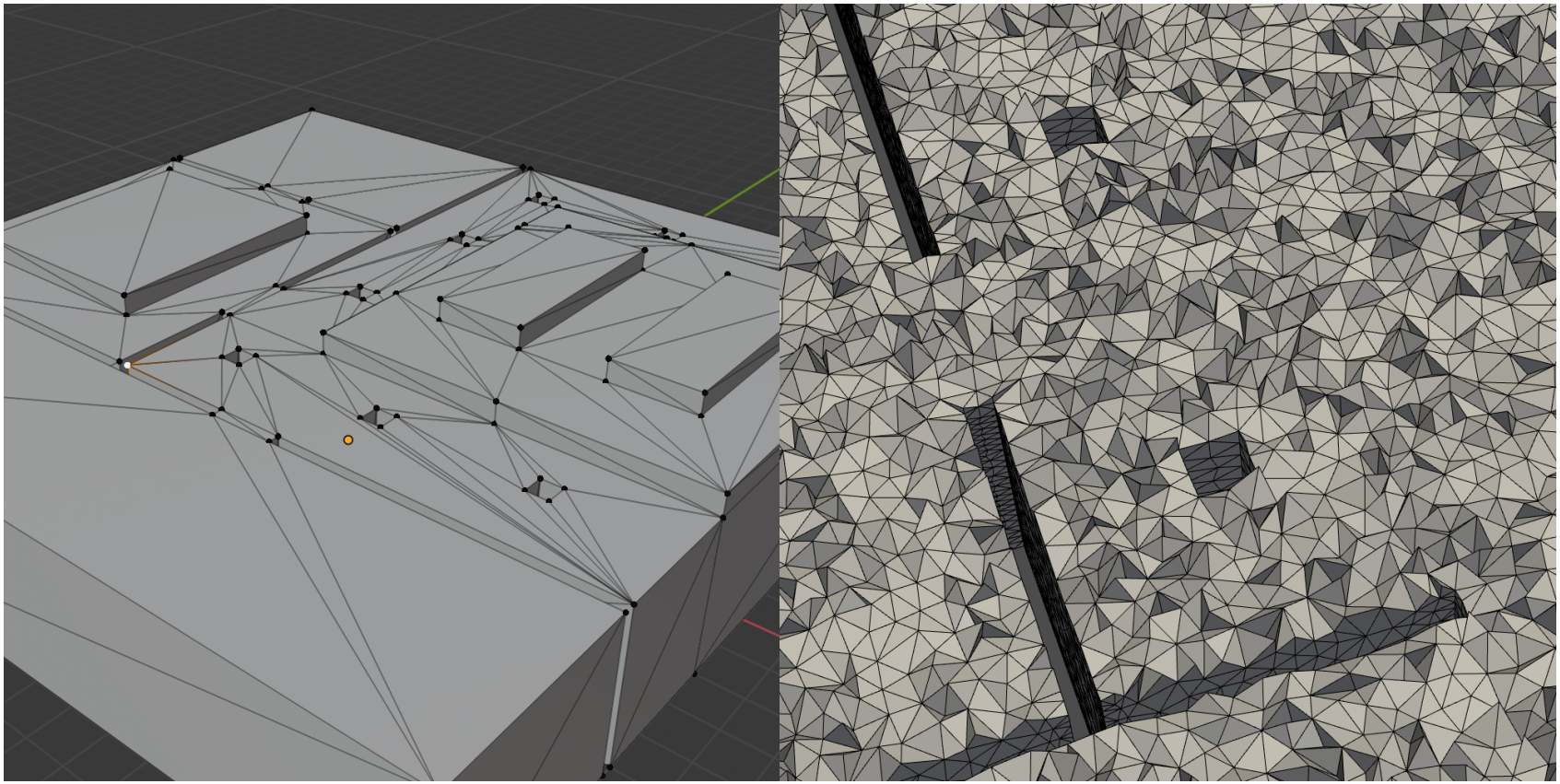
3D JMM: building test problem

Realistic test problem which includes some common architectural features, but not fully general (no small scale features, no curved surfaces, no floating obstacles, . . .):



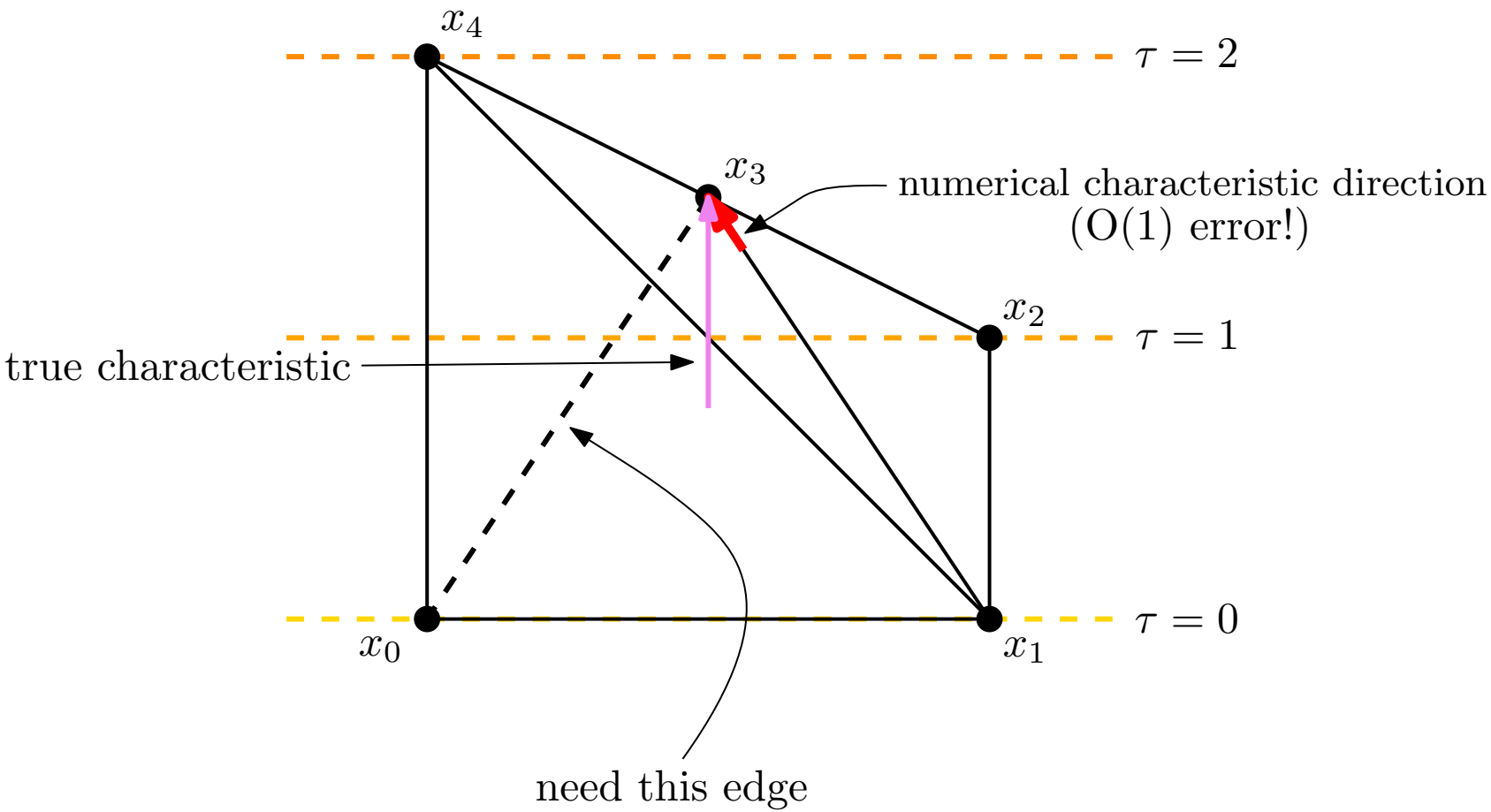
3D JMM: tetrahedron meshing

Modeled the test problem using Blender, discretized the boundary using tetgen:



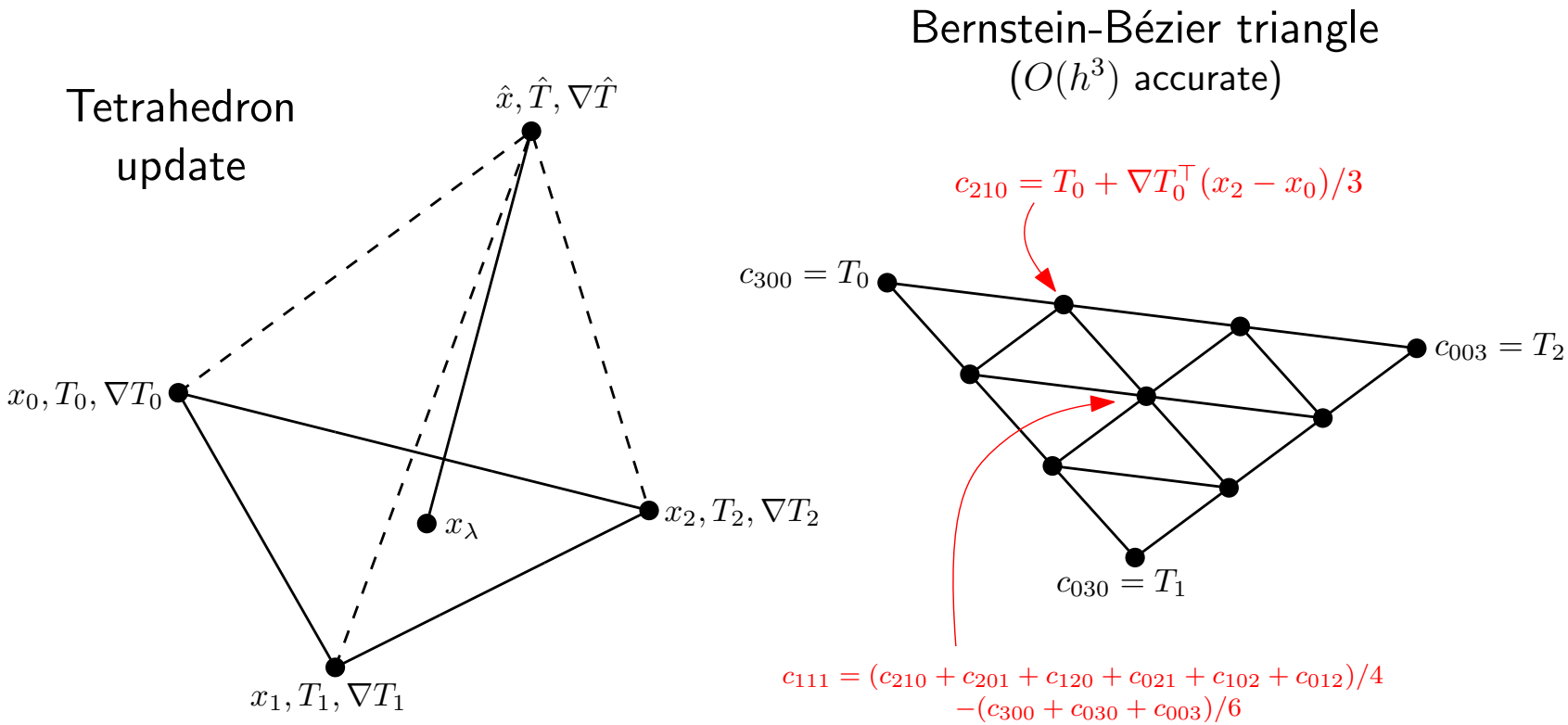
3D JMM: causality

We can't use the topology of the tetrahedron mesh to define update neighborhoods:



3D JMM: tetrahedron updates

Workhorse for this solver is a cubically accurate tetrahedron update. Since $c \equiv 1$, rays are straight lines:

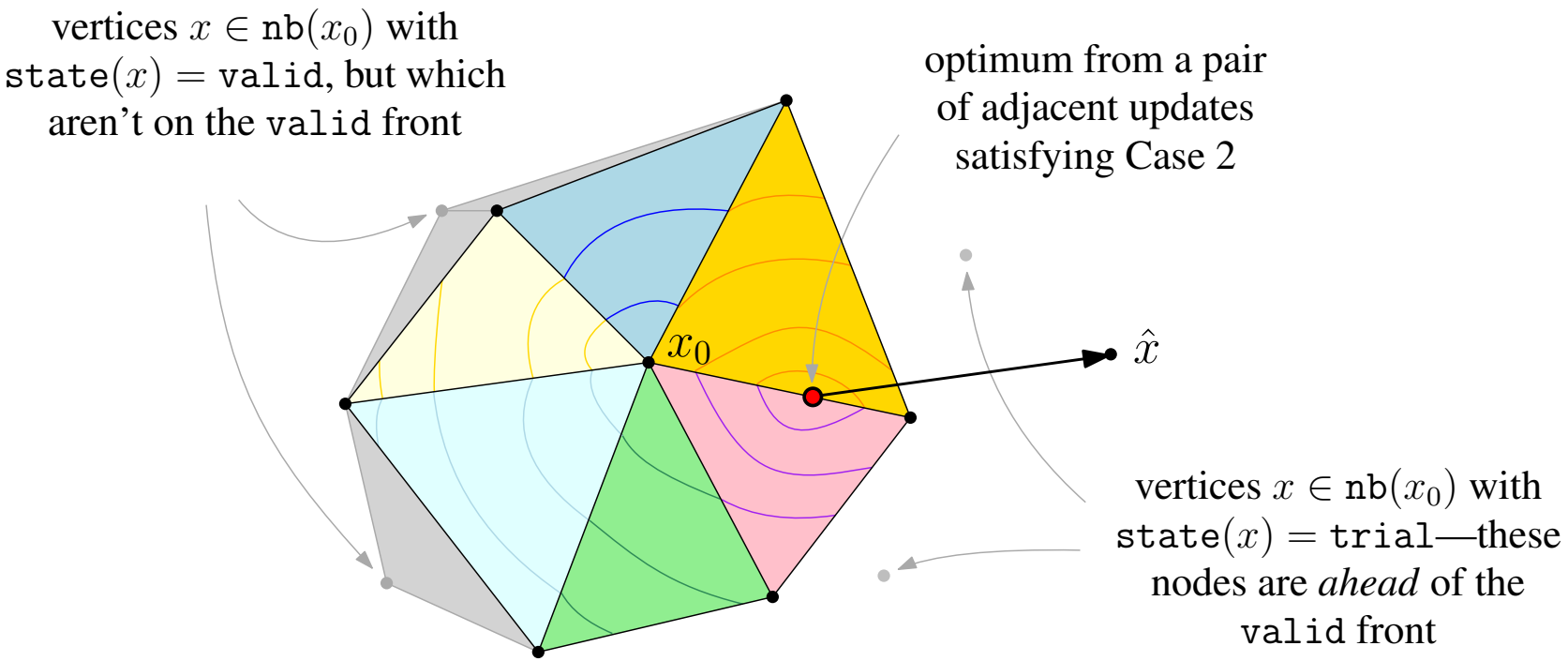


Minimize the cost function:

$$T(\hat{\boldsymbol{x}}) = \min_{\lambda} \left\{ T(\boldsymbol{x}_\lambda) + \|\hat{\boldsymbol{x}} + \boldsymbol{x}_\lambda\| \right\}$$

3D JMM: enforcing causality

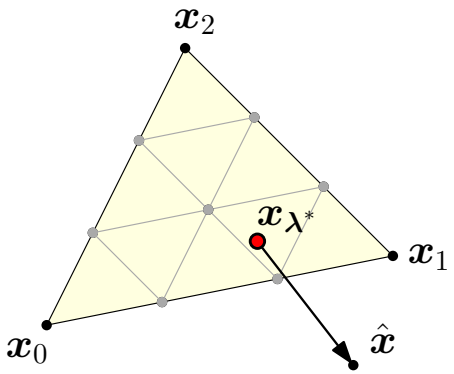
We want to find the point $\boldsymbol{x}_{\lambda^*}$ which minimizes the cost function over each `valid` triangle on the `valid` front



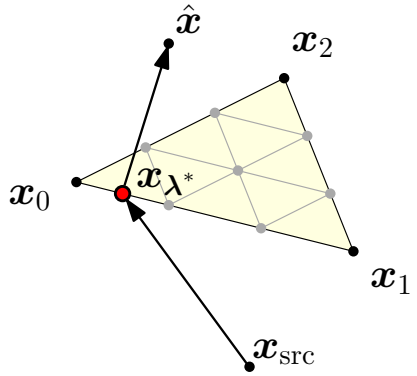
3D JMM: enforcing causality

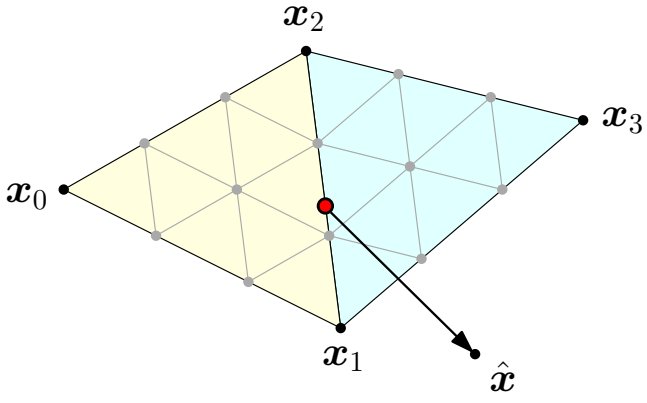
Check the Lagrange multipliers of a minimizer to see if the update is causal:

Interior point minimizer

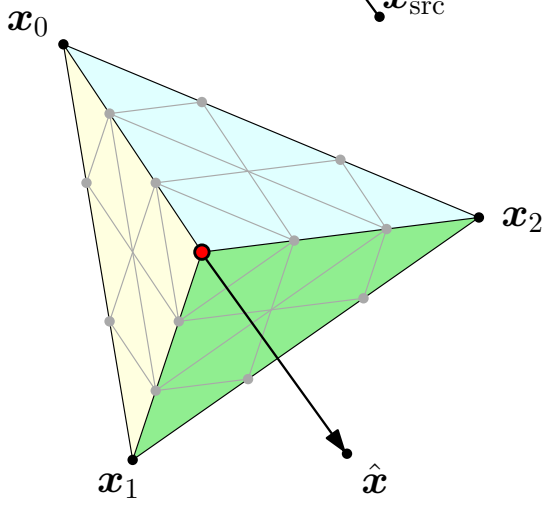


Unphysical ray



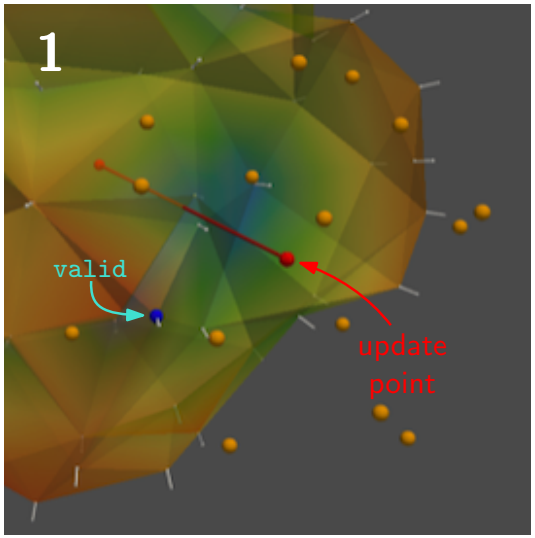


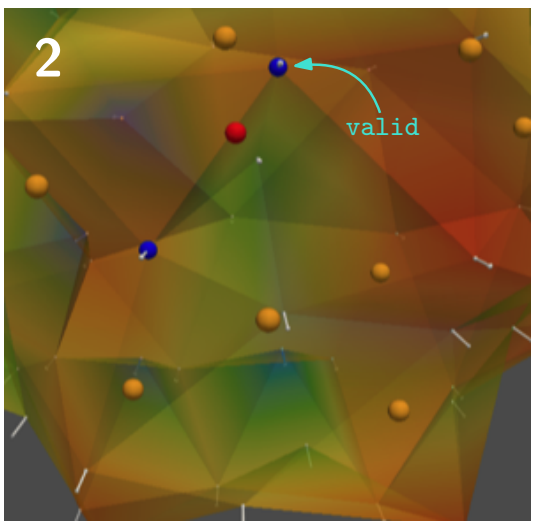
Boundary minimizer from two updates

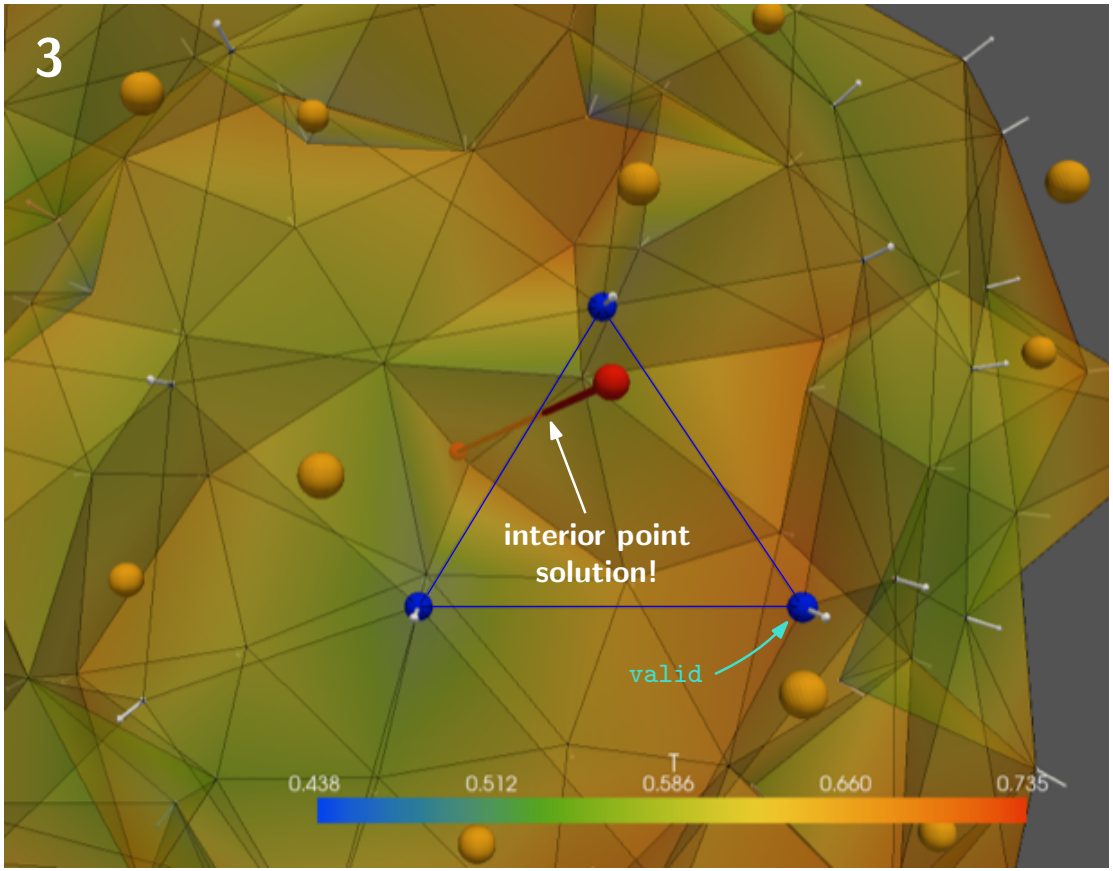


. . . and from three updates

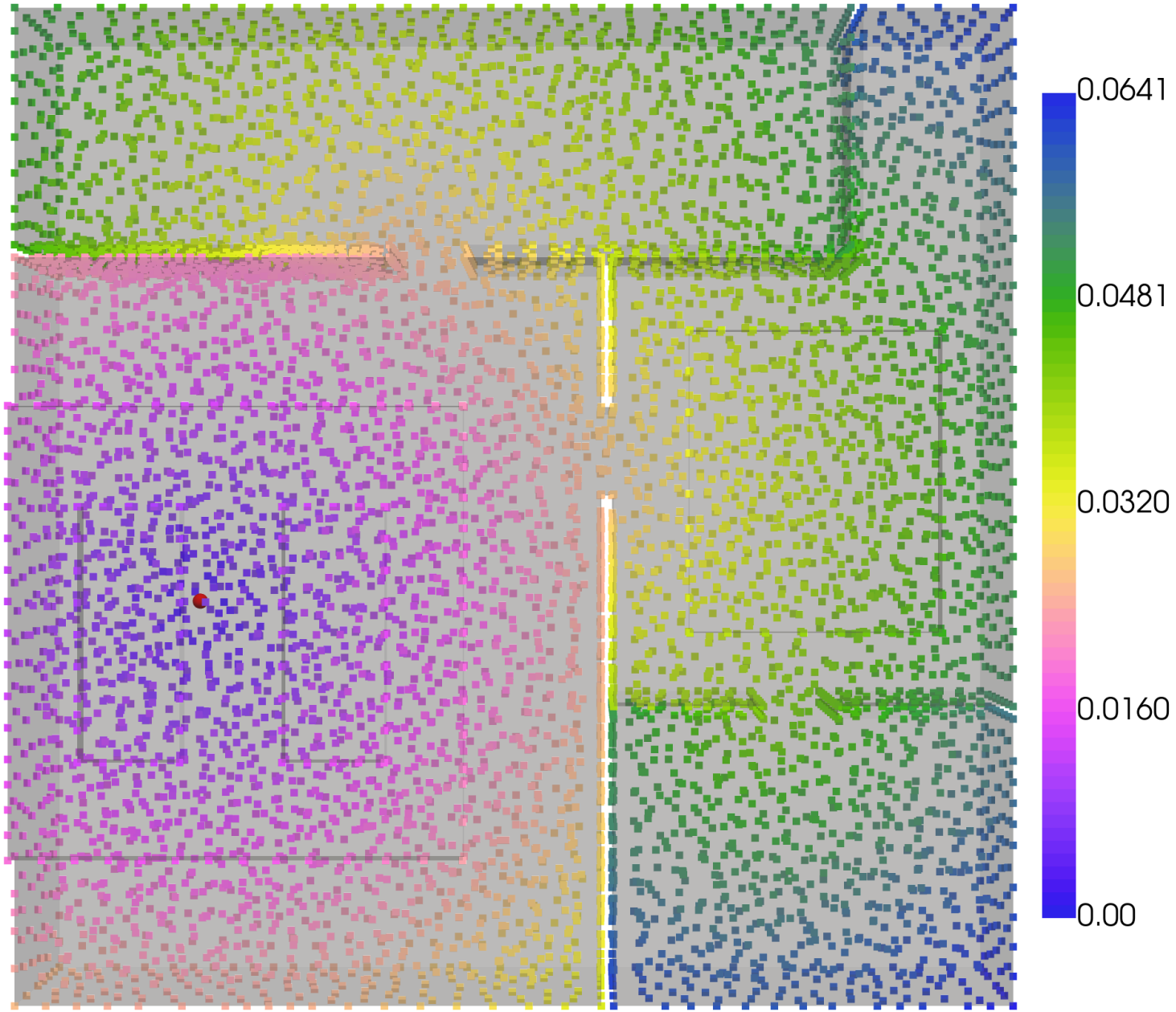
3D JMM: what does this look like?



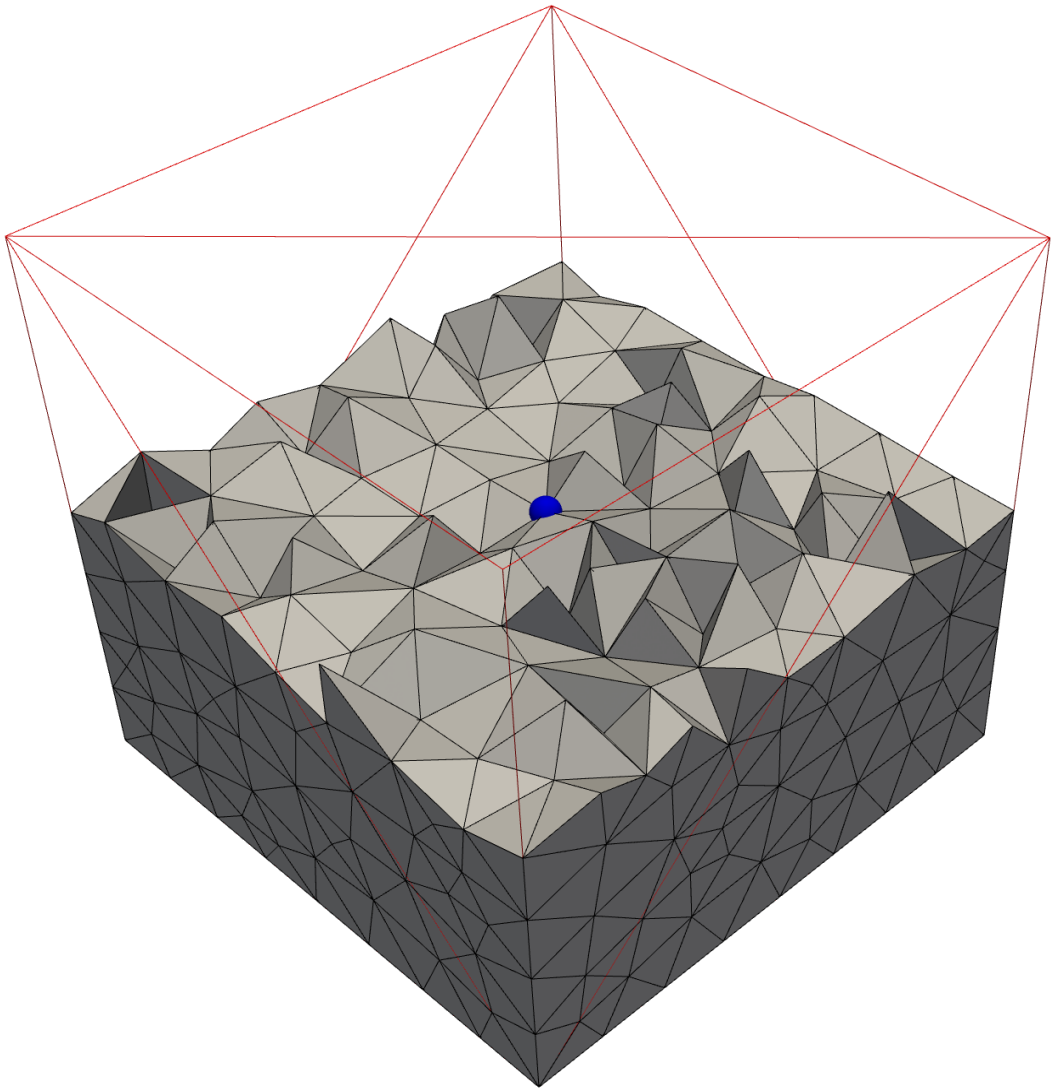




3D JMM: eikonal on the test building domain



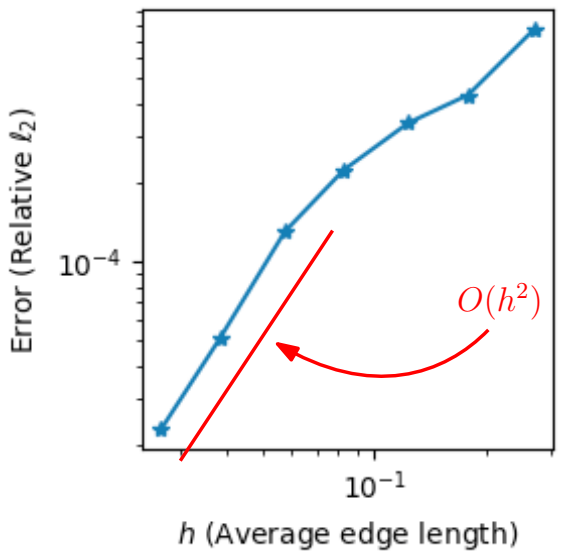
3D JMM: testing on a box domain



▲Input mesh

▲Tetrahedron mesh

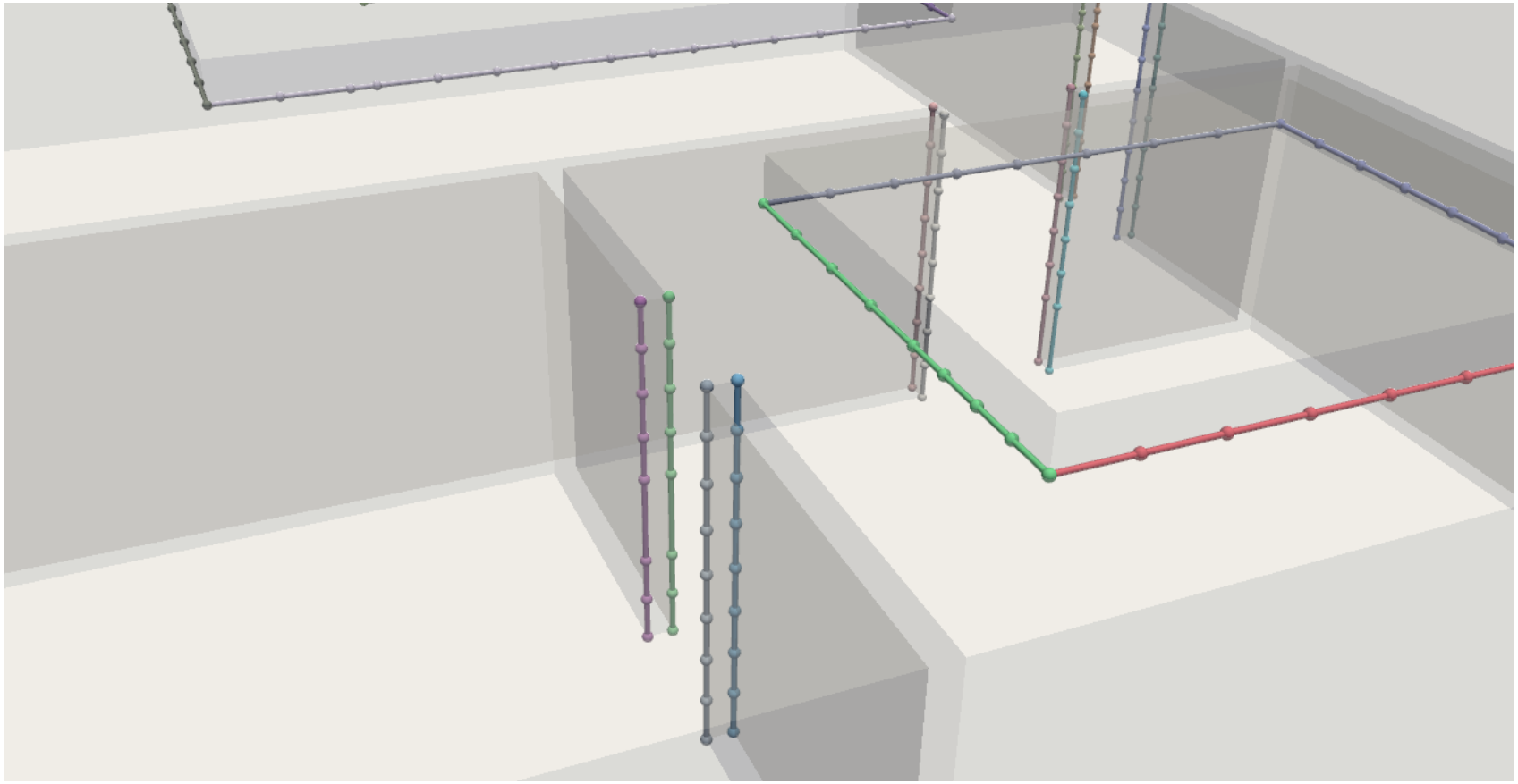
▲Point source



Multiple arrivals in 3D

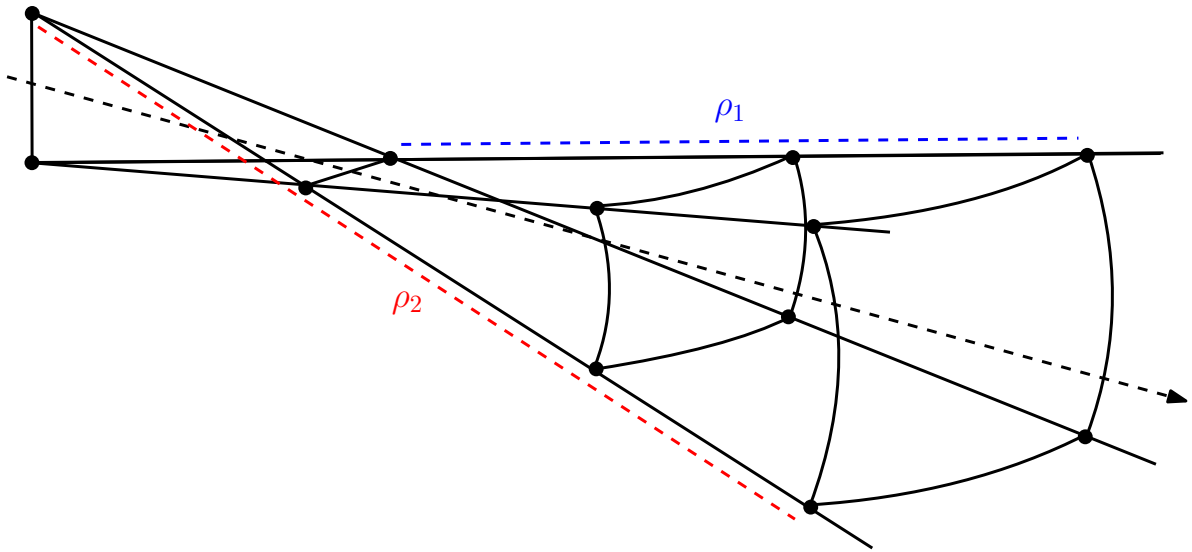
MA in 3D: reflectors and diffractors

Restrict τ_{in} to reflecting and diffracting subsets of $\partial\Omega$ and compute scattered fields:



MA in 3D: BCs for the reflected amplitude

Astigmatic pencil of rays:



The amplitude for a reflection is:

$$\alpha_{\text{out}}(\boldsymbol{x}) = R(\boldsymbol{x}_0)\sqrt{\frac{\rho_1(\boldsymbol{x}_0)\rho_2(\boldsymbol{x}_0)}{\left[\rho_1(\boldsymbol{x}_0)+\sigma(\boldsymbol{x},\boldsymbol{x}_0)\right]\left[\rho_2(\boldsymbol{x}_0)+\sigma(\boldsymbol{x};\boldsymbol{x}_0)\right]}}\,\alpha_{\text{in}}(\boldsymbol{x}_0),$$

McNamara, 1990.

MA in 3D: BCs for the diffracted amplitude

Amplitude for edge-diffracted ray is:

$$\alpha_{\text{out}}(\boldsymbol{x}) = D(\boldsymbol{x}_0)\sqrt{\frac{\rho_e(\boldsymbol{x}_0)}{\sigma(\boldsymbol{x};\boldsymbol{x}_0)\left[\rho_e(\boldsymbol{x}_0)+\sigma(\boldsymbol{x};\boldsymbol{x}_0)\right]}}\,\alpha_{\text{in}}(\boldsymbol{x}_0).$$

Complexity hidden in the UTD coefficient $D(\boldsymbol{x}_0)$:

$$D = D_1 + D_2 + R\cdot(D_3 + D_4),$$

$$D_1 = \frac{-e^{-i\pi/4}}{2n\sqrt{2\pi k}\sin\beta_0}\cot\left(\frac{\pi+\beta^-}{2n}\right)F\left(kL^{\text{in}}a^+(\beta^-)\right),$$

$$D_2 = \frac{-e^{-i\pi/4}}{2n\sqrt{2\pi k}\sin\beta_0}\cot\left(\frac{\pi-\beta^-}{2n}\right)F\left(kL^{\text{in}}a^-(\beta^-)\right),$$

$$D_3 = \frac{-e^{-i\pi/4}}{2n\sqrt{2\pi k}\sin\beta_0}\cot\left(\frac{\pi+\beta^+}{2n}\right)F\left(kL^{\text{in}}a^+(\beta^+)\right),$$

$$D_4 = \frac{-e^{-i\pi/4}}{2n\sqrt{2\pi k}\sin\beta_0}\cot\left(\frac{\pi-\beta^+}{2n}\right)F\left(kL^{\text{in}}a^-(\beta^+)\right).$$

MA in 3D: marching the Hessian

The wavefront curvature in section spanned by $\boldsymbol{v}$ and ∇T is:

$$\kappa_{\boldsymbol{v}}(\boldsymbol{x}) = -\frac{\boldsymbol{v}^\top \nabla^2 T(\boldsymbol{x}) \boldsymbol{v}}{s(\boldsymbol{x})}, \qquad \rho_{\boldsymbol{v}}(\boldsymbol{x}) = \frac{1}{\kappa_{\boldsymbol{v}}(\boldsymbol{x})}$$

The Hessian near the point source for $c \equiv 1$ is:

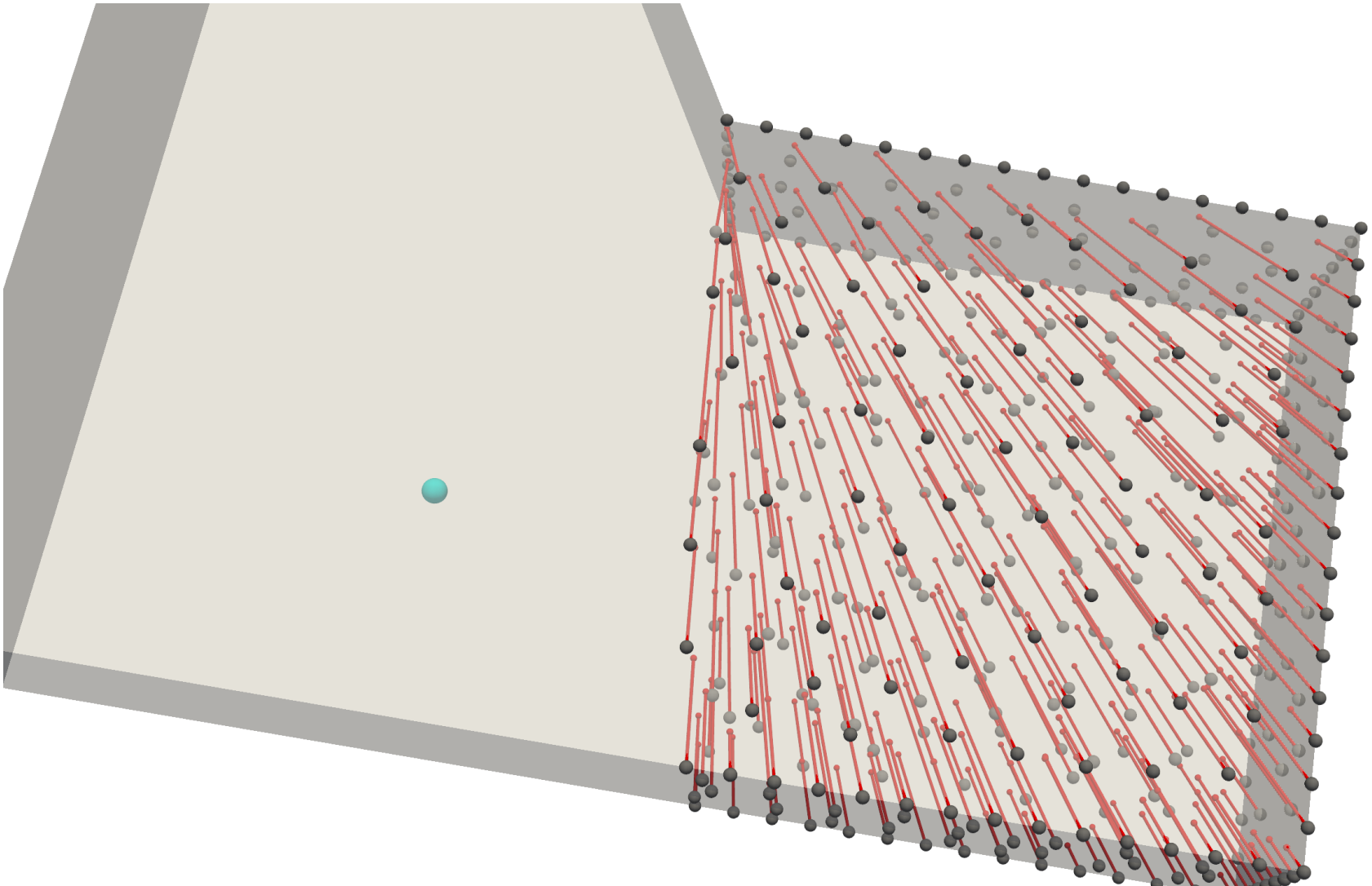
$$\nabla^2 T(\boldsymbol{x}) = \frac{1}{T(\boldsymbol{x})} \Big(\boldsymbol{I} - \nabla T(\boldsymbol{x}) \otimes \nabla T(\boldsymbol{x}) \Big).$$

Propagate the Hessian along ψ using paraxial RT:

$$\nabla^2 T(\hat{\boldsymbol{x}}) = \nabla^2 T(\boldsymbol{x}_\lambda) \Big(\boldsymbol{I} + \|\hat{\boldsymbol{x}} - \boldsymbol{x}_\lambda\| \nabla^2 T(\boldsymbol{x}_\lambda) \Big)^{-1}.$$

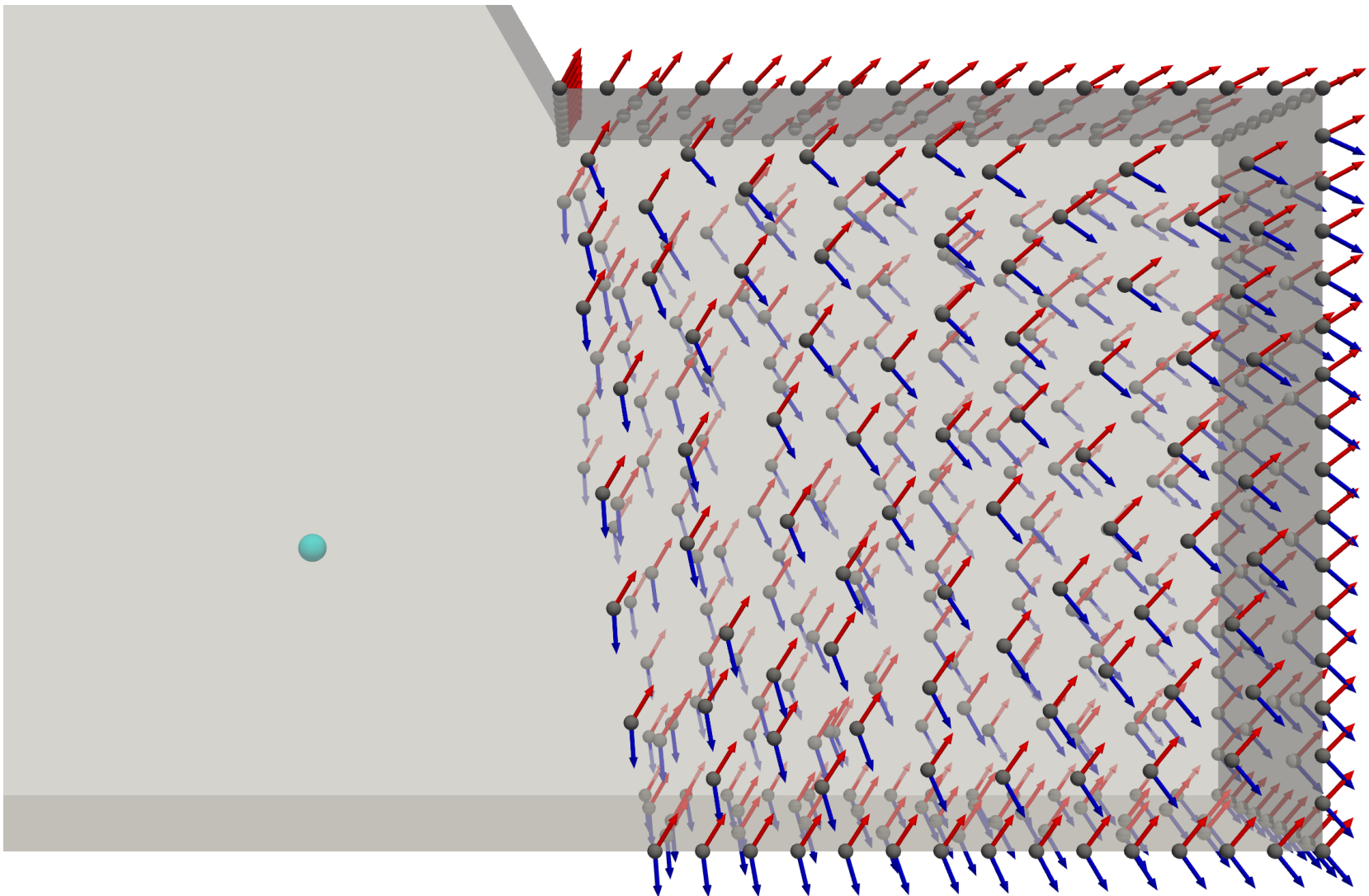
MA in 3D: evaluating BCs using dynamic programming

For $\boldsymbol{x} \in \Omega$, need to evaluate $f(\boldsymbol{x}_0)$, where $\boldsymbol{x}_0 \in \partial\Omega$ and a ray leads from $\boldsymbol{x}_0$ to $\boldsymbol{x}$. Do this after solving the eikonal equation using dynamic programming:



MA in 3D: the $\boldsymbol{t}_{\text{in}}$ and $\boldsymbol{t}_{\text{out}}$ fields

Two important vector fields are the tangent vector of the ray arriving at $\boldsymbol{x}_0$ and the ray leaving $\boldsymbol{x}_0$, which we call $\boldsymbol{t}_{\text{in}}(\boldsymbol{x})$ and $\boldsymbol{t}_{\text{out}}(\boldsymbol{x})$, respectively:



MA in 3D: shadow zones and tapering

We use $\boldsymbol{t}_{\text{in}}$ and $\boldsymbol{t}_{\text{out}}$ to taper the amplitude of points in shadow:

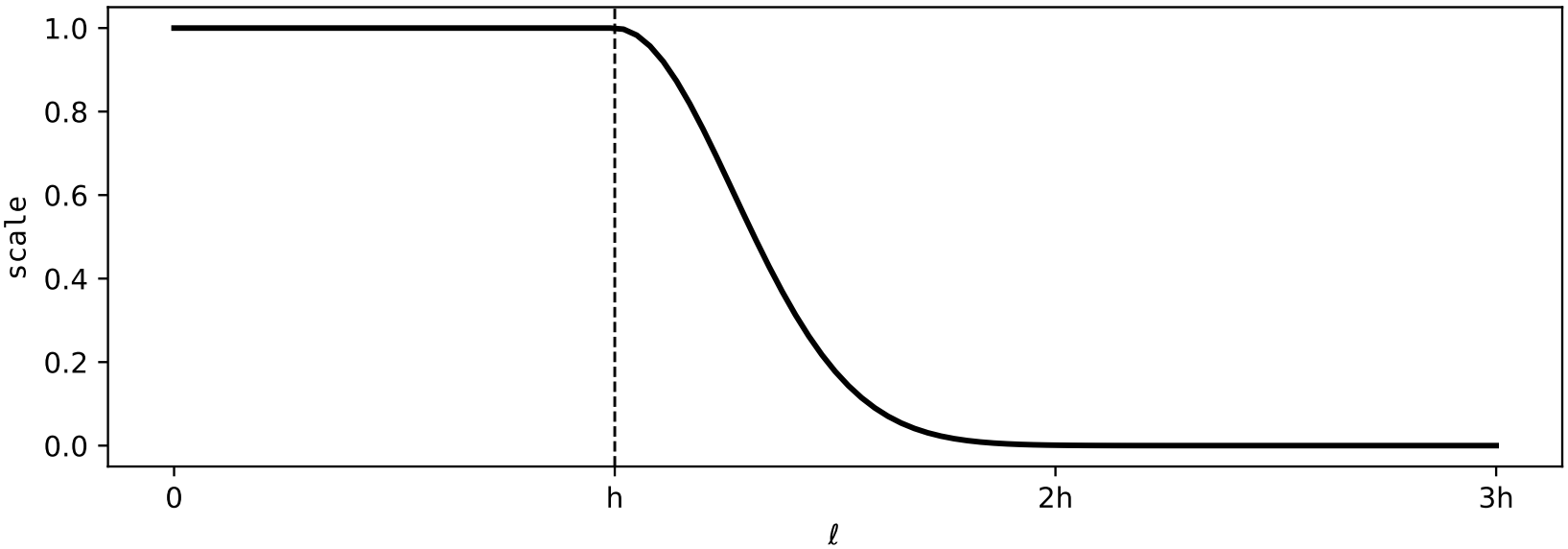
$$\ell_Z(\boldsymbol{x}) = \cos^{-1}(\boldsymbol{t}_{\text{out}}(\boldsymbol{x}) \cdot \nabla T(\boldsymbol{x}))$$

and points violating reflection and diffraction angle conditions:

$$\ell_R(\boldsymbol{x}) = \cos^{-1}\left(\boldsymbol{t}_{\text{out}}(\boldsymbol{x})(\boldsymbol{I} - 2\boldsymbol{\nu}\boldsymbol{\nu}^\top)\boldsymbol{t}_{\text{in}}(\boldsymbol{x})\right),$$

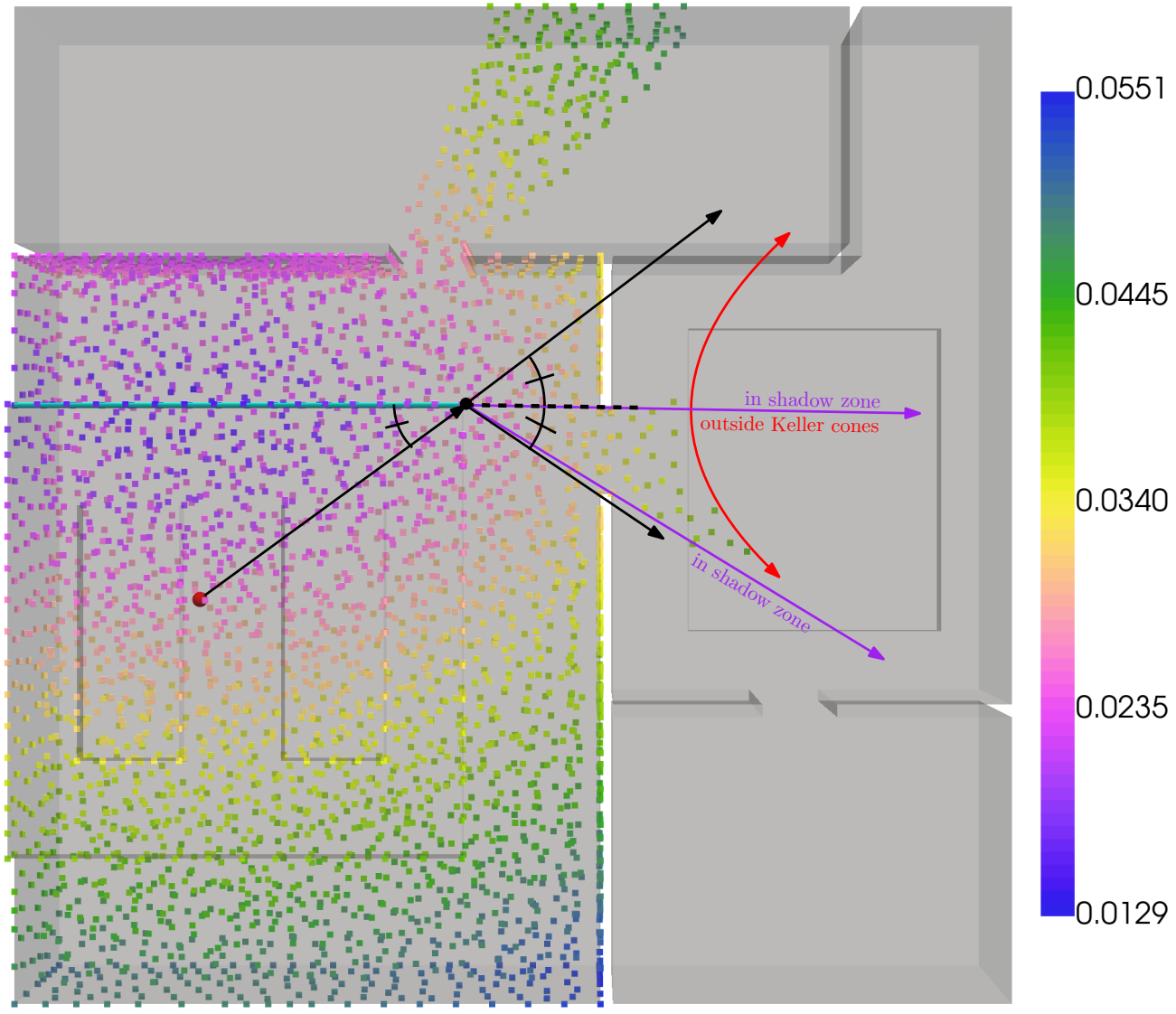
$$\ell_D(\boldsymbol{x}) = \left|\cos^{-1}\left(\boldsymbol{t}_{\text{in}}(\boldsymbol{x}) \cdot \boldsymbol{t}_{\text{e}}\right) - \cos^{-1}\left(\boldsymbol{t}_{\text{out}}(\boldsymbol{x}) \cdot \boldsymbol{t}_{\text{e}}\right)\right|$$

define `scale`(ℓ) to be Gaussian window above $\ell = O(h)$:

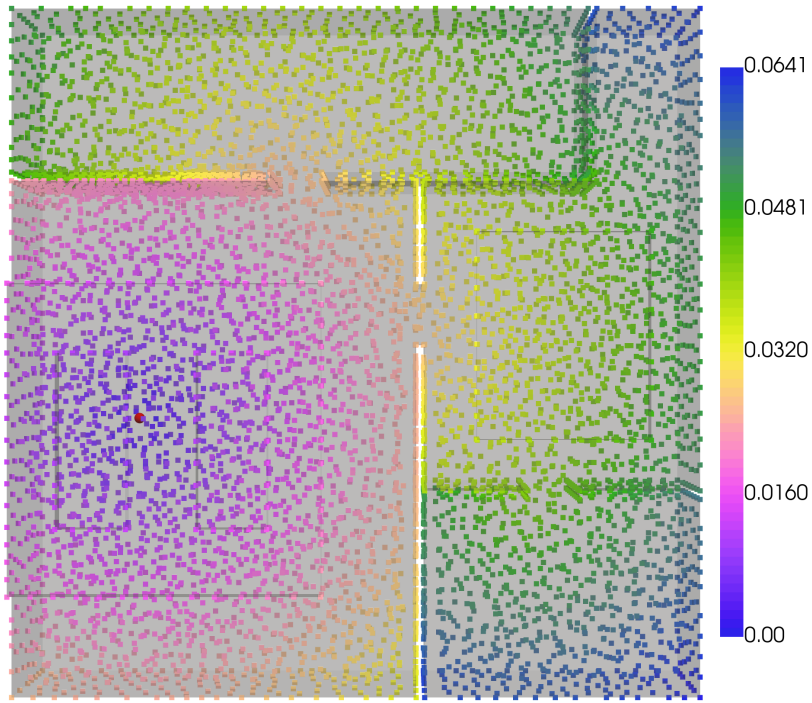


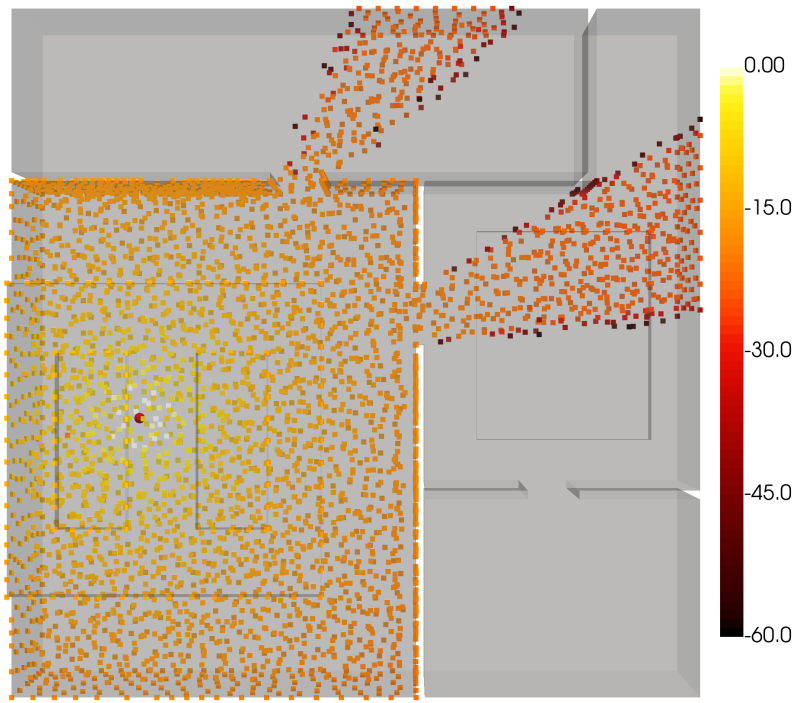
MA in 3D: shadow zones and tapering

What does this look like?

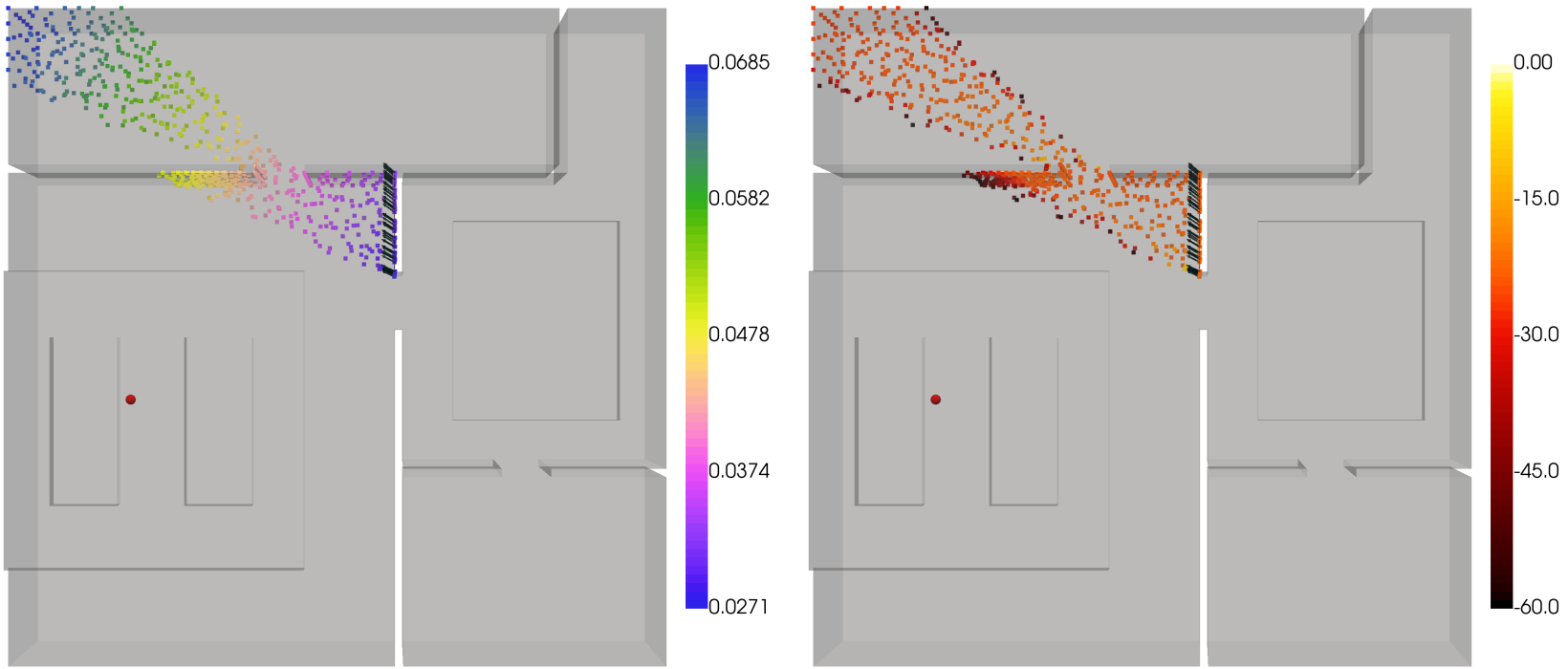


MA in 3D: building test problem

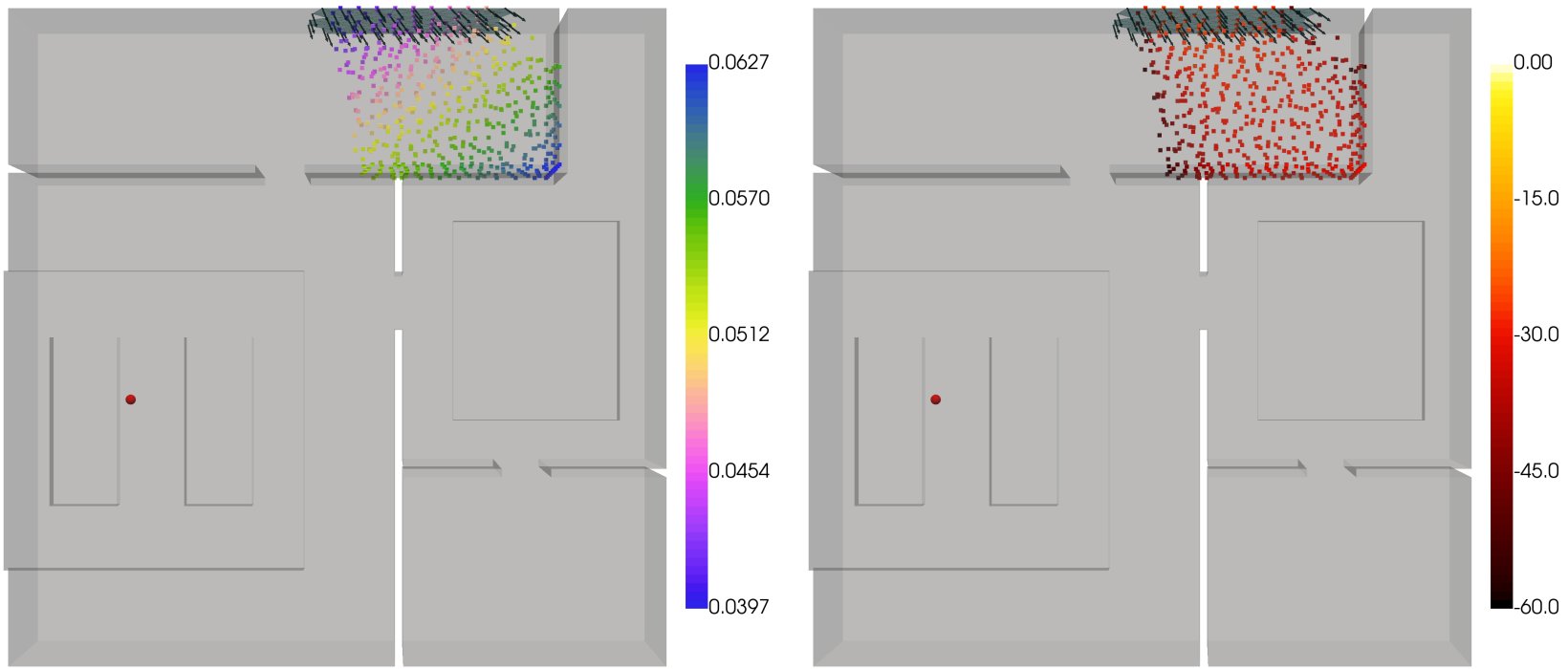




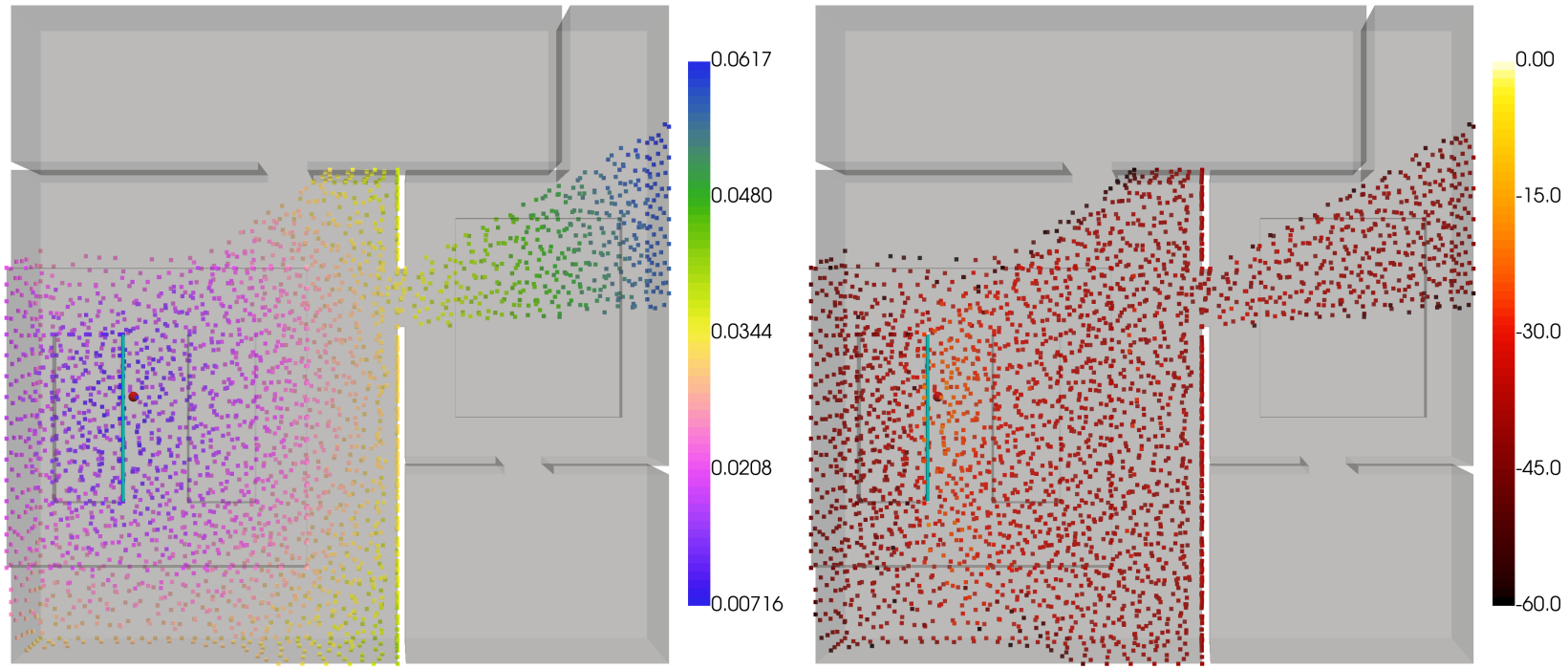
MA in 3D: building test problem



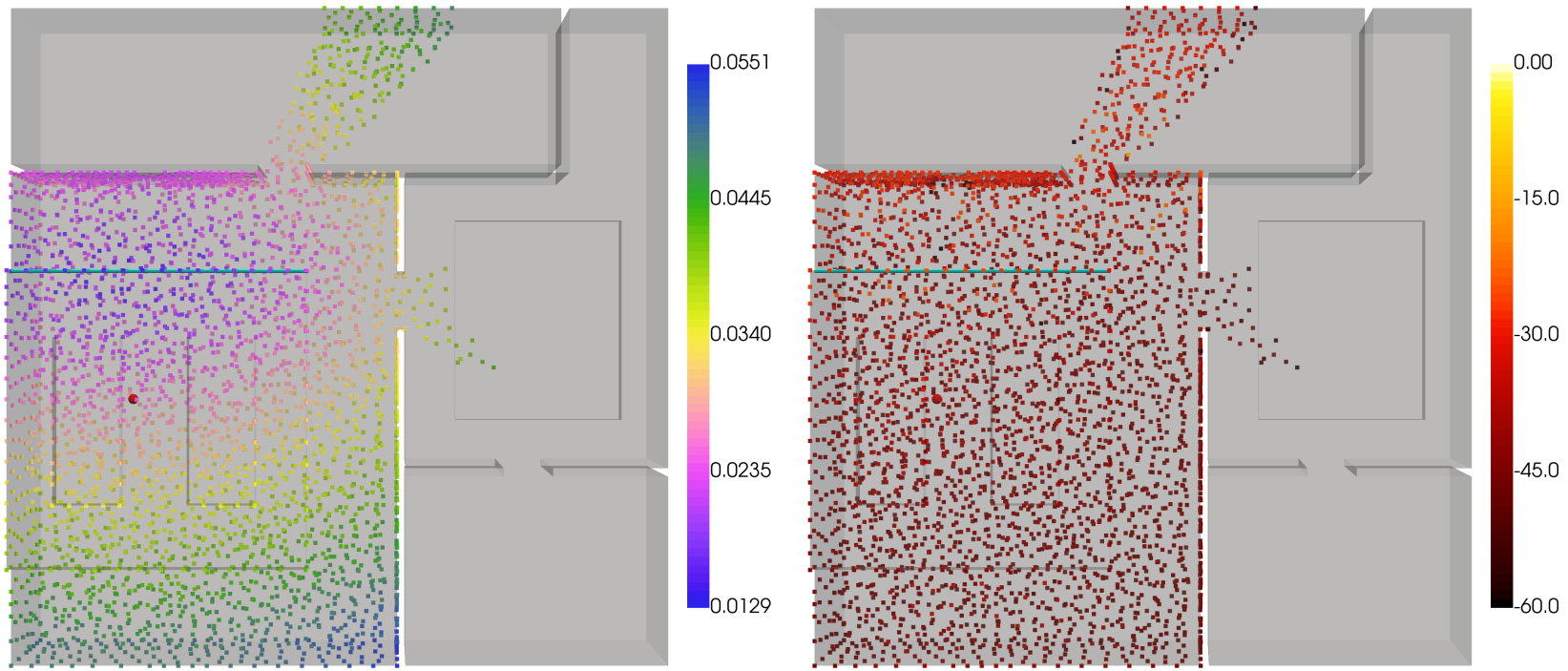
MA in 3D: building test problem



MA in 3D: building test problem



MA in 3D: building test problem



Conclusion and future work

Conclusion

OLIMs gave us intuition about how to design efficient semi-Lagrangian marchers for solving the eikonal equation, but they are only first order and not appropriate for computing the amplitude

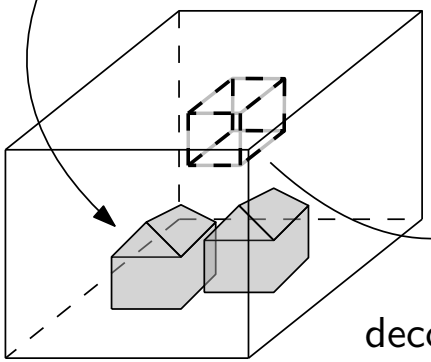
JMMs allow us to compute the eikonal with a higher order of accuracy, suitable for computing the amplitude

Realistic room acoustic modeling with multiple arrivals requires the use of an unstructured mesh. We developed a JMM which uses a novel approach to enforcing causality for computing τ on an tetrahedron mesh with this application in mind

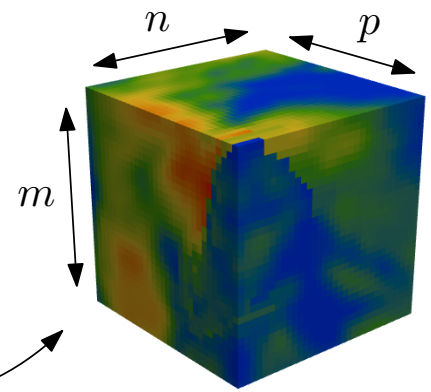
For multiple arrivals, we show how UTD can be used in a semi-Lagrangian setting by using dynamic programming, and make an initial foray into computing reflected and diffracted fields

Future work: end-to-end system

biharmonic inpainting to remove geometry



extract subvolume for each channel



collect flattened subvolumes into data matrix X

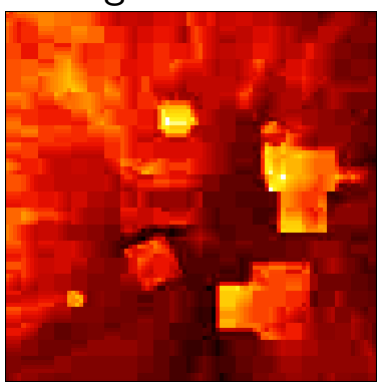
$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_c \end{bmatrix} \in \mathbb{R}^{c \times mnp}$$

$$X = U\Sigma V^\top =: UW^\top$$

$W :=$ "modes" $\in \mathbb{R}^{c \times mnp}$

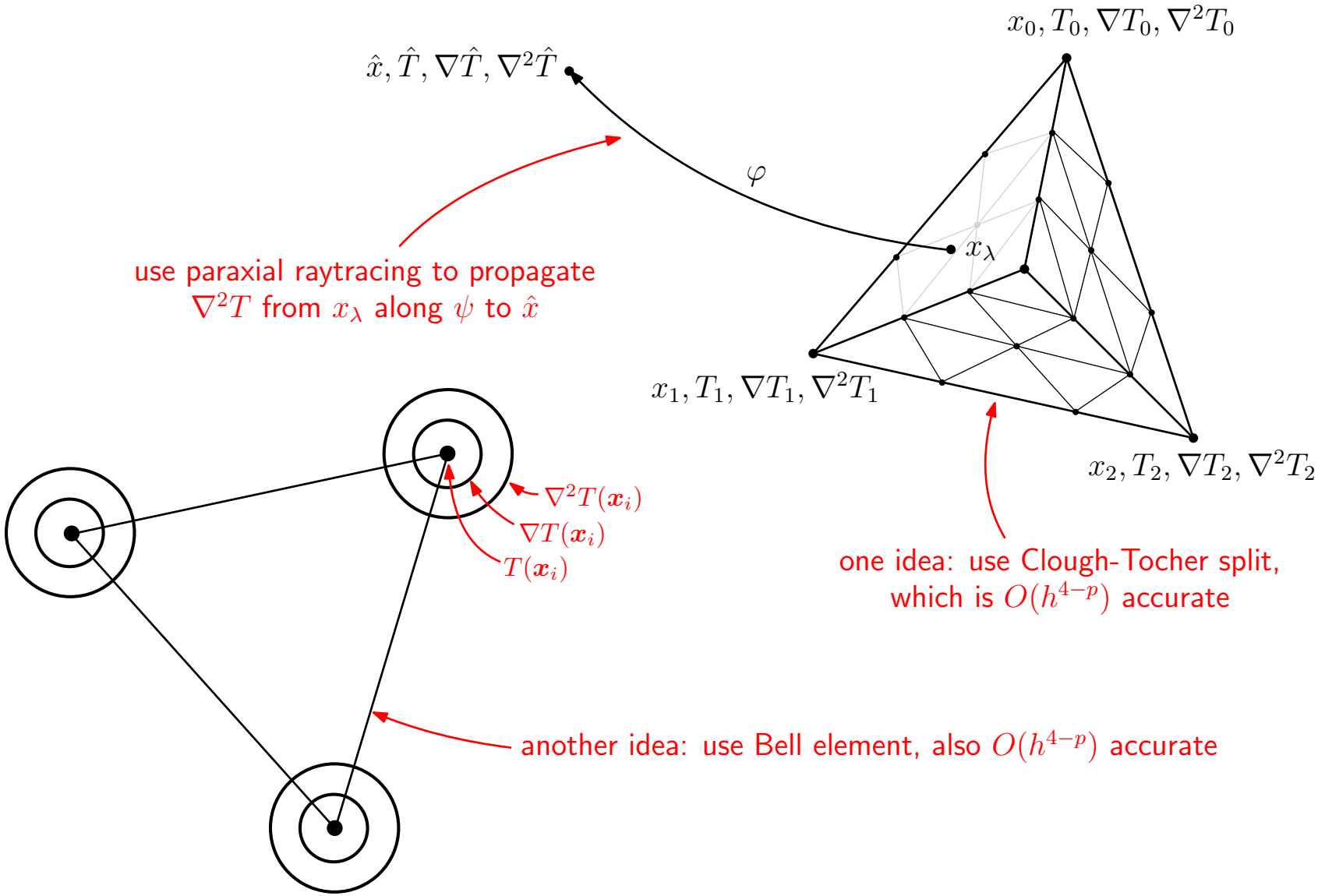
reshape each mode w_i into a $m \times n \times p$ volume, compress using lossy floating-point compression (zfp), optimize to choose compression rate

decode block by block using LRU cache



Future work: marching higher-order jets

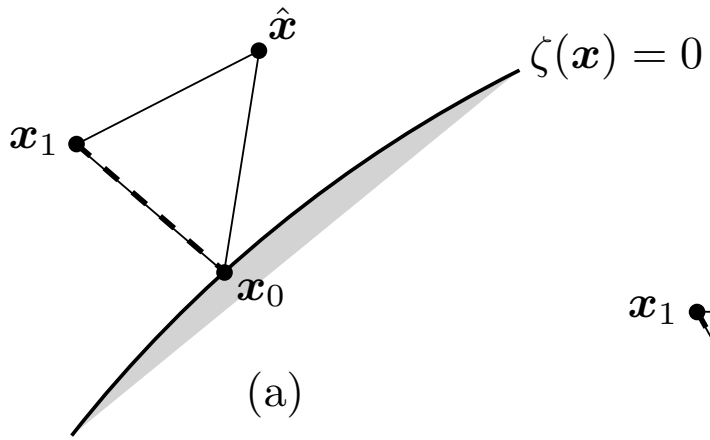
Replace 9-param triangle with **higher-order Hermite element**



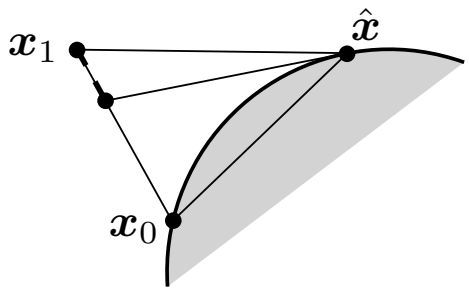
Future work: curved interfaces

Modify the optimization problems solved in the boundary layer to account for **occlusion**

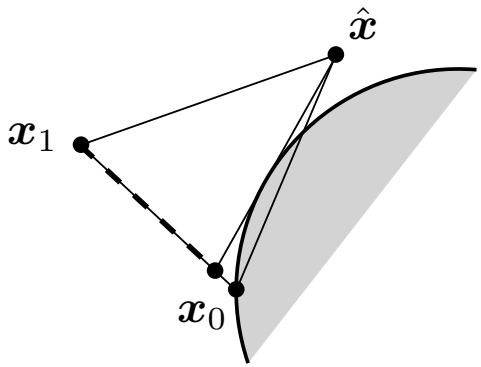
$$\begin{aligned} \text{minimize} \quad & T(\boldsymbol{x}_\lambda) + \delta T(\hat{\boldsymbol{x}}, \boldsymbol{x}_\lambda,) \\ \text{subject to} \quad & \boldsymbol{x}_\lambda \in \text{conv}(\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{x}_2) \\ & \zeta(\varphi(\sigma)) \geq 0. \end{aligned}$$



(a)



(b)

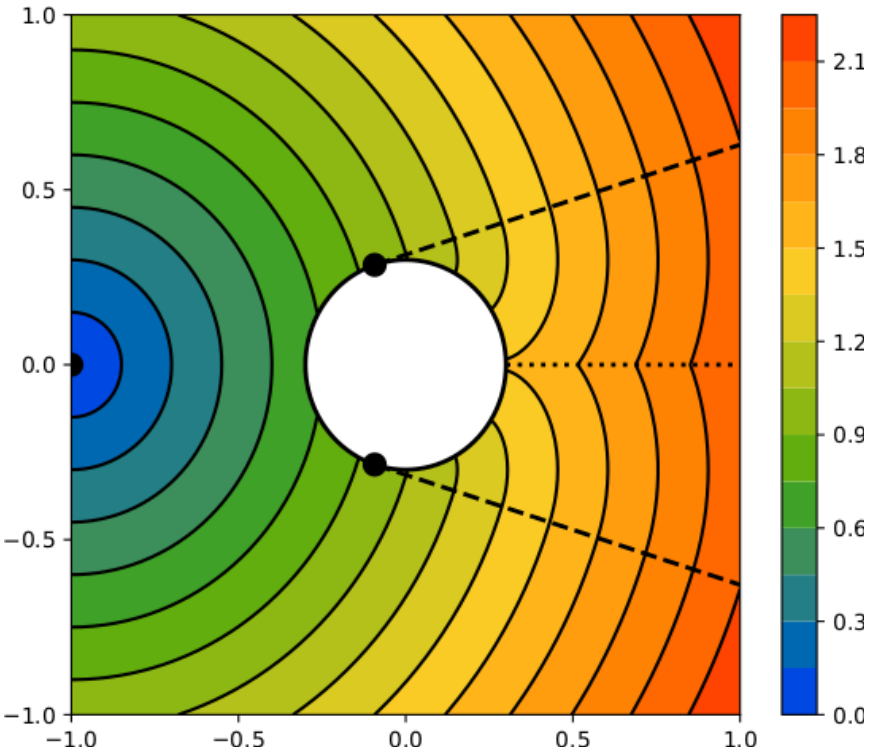


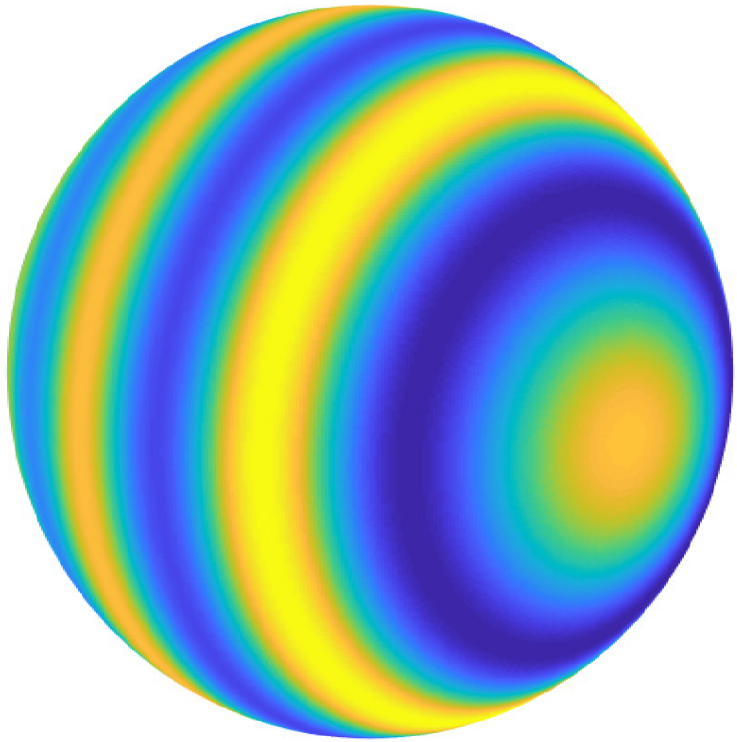
(c)

Future work: scattering from a sphere

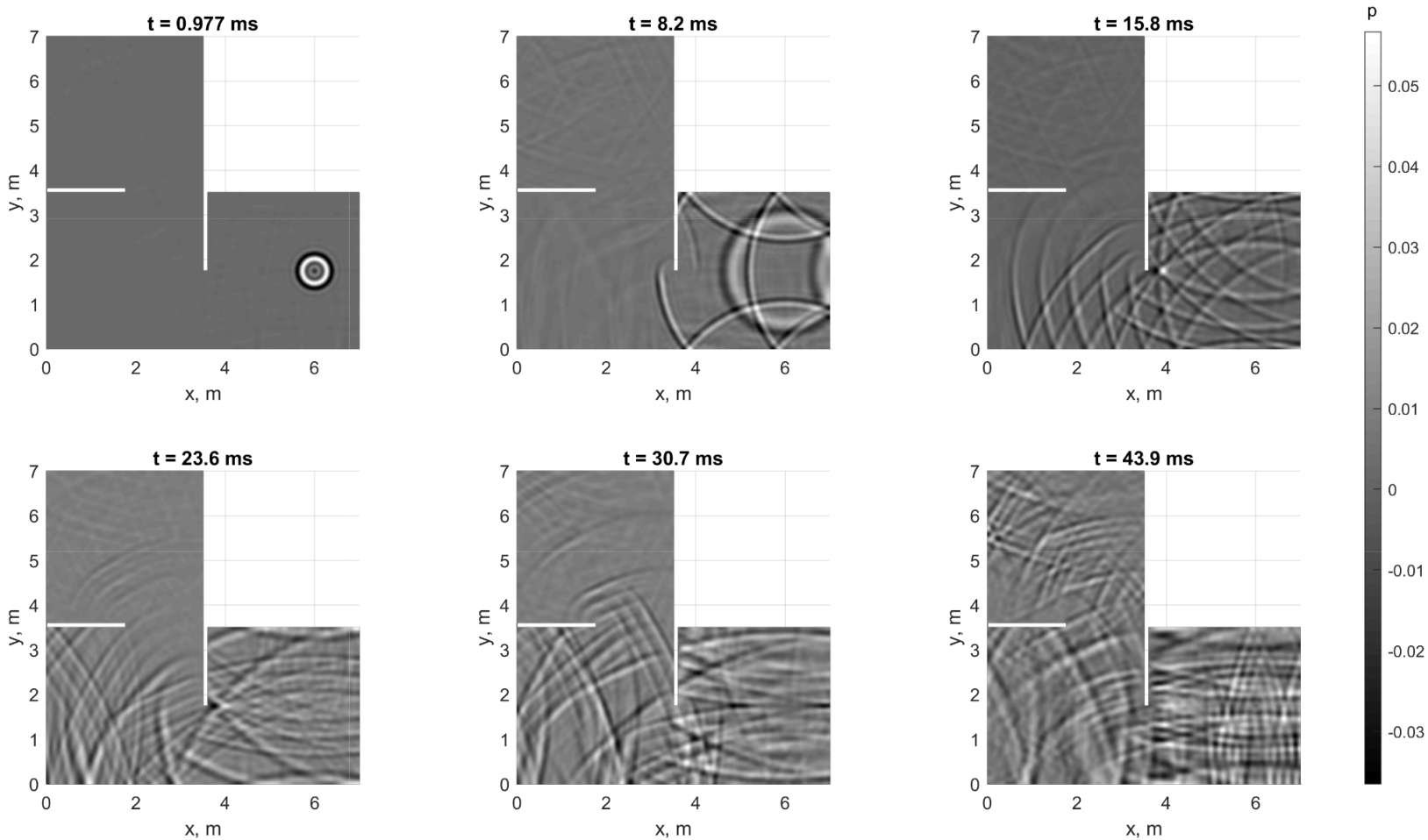
Analytic formula for sound hard scattering from a sphere:

$$P(\boldsymbol{x}) = ik \sum_{n=0}^{\infty} \sum_{m=-n}^{n} \left[j_n(kr) - \frac{j_n'(kr)}{h_n'(kr)} h_n(kr) \right] S_n^{-m}(\boldsymbol{x}_{\text{src}} - \boldsymbol{x}_0) Y_n^m(\boldsymbol{s})$$





Future work: comparison with other methods



Gumerov and Duraiswami, 2021.

Publications

Shuo Yang, **Samuel F Potter**, Maria K Cameron. *Computing the quasipotential for nongradient SDEs in 3D*, Journal of Computational Physics, 2019.

Samuel F Potter, Maria K Cameron. *Ordered line integral methods for solving the eikonal equation*, Journal of Scientific Computing, 2019.

Samuel F Potter, Maria K Cameron. *Jet marching methods for solving the eikonal equation*, in review.